Highlights

Random Matrix Filtering and Planar Financial Networks for Volatility Forecasting in the Brazilian Stock Market

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- RMT provides a spectral decomposition of market-wide, group-level and noise-compatible components in Brazilian equities.
- PMFG filtering highlights localized dependency patterns consistent with sectoral organization.
- Within-sector correlations are larger than between-sector correlations on average.
- Network features provide limited but nonzero incremental evidence for volatility forecasting.
- GARCH and HAR-RV remain strong benchmarks for network-augmented machine learning.

Random Matrix Filtering and Planar Financial Networks for Volatility Forecasting in the Brazilian Stock Market

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ABSTRACT

Financial markets are complex systems in which collective fluctuations, sectoral organization and sampling noise coexist in empirical correlation matrices. This paper develops an econophysics pipeline for Brazilian equities traded on B3 from 1998 to 2025 and evaluates whether the resulting dependency structure contains information for volatility forecasting. Using adjusted daily prices, we compute log returns for a historical universe of 58 equities and combine stylized-fact diagnostics, Random Matrix Theory (RMT), hierarchical clustering, Minimum Spanning Trees (MSTs), Planar Maximally Filtered Graphs (PMFGs) and realized-volatility forecasting models. Using the Marčenko–Pastur benchmark, the empirical spectrum is decomposed into a dominant market mode, candidate group/sector modes and a residual noise-compatible component; five empirical eigenvalues lie above the Marčenko–Pastur upper bound, and the largest eigenvalue accounts for about 37% of the trace of the correlation matrix. Filtered MST and PMFG representations show that removing the market mode reduces average edge correlations while retaining patterns consistent with localized sectoral organization. The group-mode PMFG has lower mean edge correlation but higher clustering than the original network, suggesting stronger local clustering in the mesoscopic representation. Forecasting experiments for PETR4, VALE3 and BBDC4 indicate that GARCH(1,1) and HAR-RV remain strong benchmarks, especially at short horizons, while machine-learning models with PMFG-derived features provide competitive and incremental evidence rather than uniformly dominating classical models. The results support the view that RMT-filtered topology is useful for interpreting market structure and may complement volatility-risk modelling in emerging equity markets.

1. Introduction

Financial markets are often studied as complex systems in which heterogeneous agents, institutional constraints, liquidity cycles, information flows and feedback mechanisms interact across multiple time scales [24, 11]. Even when the empirical object is reduced to daily asset returns, market variation should not be represented only by the marginal behaviour of each series. In practice, the variation of financial returns available in a market must also be described through the structure of relations among assets, which is summarized by their correlation matrix. This matrix reflects a mixture of collective market motion, sectoral organization, idiosyncratic shocks, crisis amplification and sampling noise. A central task of empirical econophysics is therefore to distinguish interpretable dependency structure from fluctuations that are compatible with random estimation effects [33].

This distinction is relevant because financial risk is inherently multivariate. The volatility of an individual asset matters, but portfolio risk, contagion, diversification and systemic vulnerability depend on how assets co-move [3]. During calm periods, correlations may appear moderate and dispersed. During stress periods, correlations often increase, reducing diversification benefits and lowering the effective dimensionality of the market [18, 44]. A purely univariate analysis cannot represent this collective component, whereas an unfiltered analysis of the full empirical correlation matrix may confuse persistent structure with noise.

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Emerging equity markets are particularly relevant in this context. They combine global risk channels with local institutional, macroeconomic and liquidity conditions. Brazil provides a suitable empirical setting because B3 includes globally exposed commodity companies, large financial institutions, regulated utilities, consumer cyclicals, industrial firms, telecommunications firms and real-estate companies. These assets share exposure to domestic monetary policy, exchange-rate dynamics, country risk and global liquidity, but they also respond to distinct sectoral fundamentals. These characteristics suggest that the empirical dependency matrix may combine a dominant market component, group or sectoral components and a large residual component. Despite these empirical features, the Brazilian equity market remains relatively underexplored in the complex-systems literature. Previous studies have addressed Brazilian interbank networks [39], global financial system linkages [34], agrarian cross-correlations [38], multiscale networks [29] and equity-market topology [22], but a broad B3 equity pipeline combining Random Matrix Theory filtering, planar financial networks and volatility forecasting remains insufficiently developed.

The empirical difficulty is that correlation matrices are noisy, especially when the number of assets is not negligible relative to the number of observations. Random Matrix Theory (RMT) provides a benchmark for this problem by comparing the empirical eigenspectrum with the spectrum expected from random correlation matrices [25, 20, 30]. Eigenvalues lying inside the Marčenko–Pastur band are interpreted as compatible with random sampling noise under the benchmark model, whereas eigenvalues outside the band indicate candidate collective modes that deserve further interpretation. Subsequent work has extended RMT-based methods for cleaning large covariance matrices [8]. RMT is not used here as a mechanical classifier of economic sectors, but as an intermediate layer for reconstructing broad market, group-level and noise-compatible components before graph construction.

Network filtering provides a complementary representation of the same dependency system. By transforming correlations into distances, financial networks retain selected dependency links among assets and make it possible to inspect market backbones, local clusters and central nodes. The Minimum Spanning Tree (MST) emphasizes the sparsest connected structure [23], while the Planar Maximally Filtered Graph (PMFG) retains a denser planar representation with richer local connectivity [40, 41]. These constructions allow one to ask whether strong dependencies are organized by sectors, whether central nodes change after market-mode removal and whether the mesoscopic topology of the market contains interpretable information beyond average correlation. Filtered networks are also relevant for risk analysis, where peripheral assets can offer better diversification properties than central hubs [32, 2].

This paper develops an integrated econophysics pipeline for Brazilian equities traded on B3. The analysis proceeds through three stages. The first stage documents stylized facts, static correlations, sectoral correlation differences and rolling market synchronization. The second stage applies RMT to decompose the empirical correlation matrix into market, group/sector and noise components, followed by hierarchical clustering and ordered heatmaps. The third stage converts original and filtered matrices into MST and PMFG networks, studies hub reorganization and aggregates dependencies at the subsector level. A volatility forecasting experiment then evaluates whether the extracted features provide incremental predictive information relative to established econometric benchmarks [4, 12, 16].

The contribution of this paper is fourfold:

1. It builds a reproducible econophysics pipeline for B3 equities using adjusted daily prices from 1998 to 2025, including return construction, stylized-fact diagnostics, correlation analysis, sectoral comparison and rolling synchronization.
2. It applies Random Matrix Theory to a 58-asset historical core universe and reconstructs market, group/sector and noise-compatible components of the empirical correlation matrix, providing a spectral interpretation of Brazilian equity dependencies.
3. It compares MST and PMFG networks built from original and RMT-filtered matrices, with emphasis on hub reorganization, clique structure, clustering, sectoral retention and subsector-level aggregation.
4. It evaluates whether RMT- and network-derived features provide incremental predictive information for realized-volatility forecasting relative to GARCH(1,1), HAR-RV, EWMA and classical volatility predictors, using a strict temporal split and nested feature-set design.

The article is organized as follows. Section 2 reviews financial dependencies, Random Matrix Theory, filtered financial networks and volatility forecasting. Section 3 describes the data and asset universe. Section 4 presents the empirical methodology and the mathematical definitions used throughout the paper. Sections 5–10 report the empirical results. Sections 11 and 12 discuss interpretation and limitations, and Section 13 presents the final considerations.

2. Related work

2.1. Financial dependencies and econophysics

Econophysics studies financial markets using concepts and methods from statistical physics, nonlinear dynamics, complex systems and network science. A central empirical observation is that financial returns exhibit stylized facts that are difficult to reconcile with independent Gaussian models. Heavy tails, volatility clustering, slow decay in absolute-return autocorrelations, intermittent bursts of activity and collective co-movement patterns are recurrent findings across asset classes, countries and sampling frequencies [24, 11]. These empirical regularities motivate methods that focus not only on the marginal distribution of returns, but also on dependency, synchronization and topology.

Financial dependency analysis lies at the intersection of econophysics and financial economics. In financial economics, dependence is central to portfolio risk, diversification, contagion, factor structure and systemic vulnerability. In econophysics, the same dependence is often represented through correlation matrices, eigenspectra, filtered graphs and hierarchical structures. Raddant and Di Matteo [33] emphasize that correlations, causality measures, volatility spillovers and networks provide complementary views of financial systems. The present paper follows this perspective: the empirical goal is not merely to estimate pairwise correlations, but to decompose the correlation matrix, filter its noisy components, represent the remaining structure as networks and evaluate whether the extracted topology has incremental forecasting relevance.

Cross-correlation dynamics during crises have received particular attention because market stress often increases synchronization across assets. Zheng et al. [44] showed that changes in cross-correlation structure can serve as early indicators of systemic instability. Billio et al. [3] proposed econometric measures of connectedness and systemic risk in financial sectors. Junior and Franca [18] documented that correlations among global stock indices tend to increase during crises, reducing the effective dimensionality of international markets. These findings motivate the rolling-correlation and crisis-synchronization analysis undertaken in this paper, where market-wide co-movement is treated as an empirical object rather than as a nuisance.

The relevance of these studies for the present manuscript is twofold. First, they show that financial dependence is dynamic and crisis-sensitive. Second, they indicate that a single unfiltered correlation matrix may conceal different layers of structure: a broad market component, group or sectoral organization and residual noise. This motivates the use of Random Matrix Theory and filtered networks as complementary tools for separating and interpreting these layers.

2.2. Random Matrix Theory in financial correlations

Random Matrix Theory entered empirical finance as a way to separate information from noise in large correlation matrices. Laloux et al. [20, 21] showed that a substantial part of the eigenspectrum of empirical financial correlation matrices can resemble the spectrum predicted for random matrices. The relevant benchmark is the Marčenko–Pastur distribution [25], which provides bounds for the eigenvalues of a random correlation matrix estimated from independent standardized time series. Plerou et al. [30] further studied eigenvalues and eigenvectors outside the random band and interpreted the largest eigenvalue as a market-wide mode.

The intuition is that the empirical correlation matrix contains both structured and noisy components. If N assets are observed over T periods, the ratio $Q = T/N$ determines the width of the random-matrix noise band. Eigenvalues inside this band are treated as compatible with sampling noise, whereas eigenvalues above the upper edge are interpreted as candidates for collective factors. This interpretation is particularly useful in financial markets because broad market co-movement can dominate the first eigenvalue, while weaker sectoral or group modes may appear in subsequent eigenvectors.

Subsequent work extended RMT-based methods for covariance and correlation cleaning. Bun et al. [8] reviewed and unified several approaches for cleaning large covariance matrices using random matrix theory, connecting theoretical results with practical financial applications. Guhr et al. [15] provide a broader mathematical review of random matrix theory and its applications in complex systems. In finance, RMT has been used not only for noise filtering, but also for portfolio optimization, factor identification and the study of collective market behaviour.

The RMT interpretation used in this manuscript is deliberately cautious. The largest eigenvalue is interpreted as a market mode because it captures broad positive co-movement across many assets. Eigenvalues above the Marčenko–Pastur upper edge, excluding the first one, are interpreted as group or sector candidates. Eigenvalues inside the random band are treated as noisy residual components. This terminology is a filtering convention, not a claim that every eigenvector has a unique economic label. The contribution of the present paper is to use this spectral separation as the input for network construction and volatility forecasting in the Brazilian equity market.

2.3. Financial networks: MST, PMFG and filtered graphs

Financial networks transform a dependency matrix into a graph representation. The usual approach begins by converting correlations into distances, so that strongly correlated assets are close in the induced metric space. Mantegna [23] introduced the Minimum Spanning Tree (MST) as a way to reveal hierarchical organization in financial markets. The MST retains the connected tree with minimum total distance and therefore contains exactly $N - 1$ edges for N assets. Its main advantage is interpretability: it provides a sparse backbone of the strongest dependencies. Its main limitation is that it removes cycles and many potentially informative local connections.

Subsequent studies applied MST-based methods to study market taxonomy, instability and dynamics. Onnela et al. [27] analysed the evolution of MST-based market correlations and their implications for portfolio structure. Bonanno et al. [5] extended the network approach to multiple world markets, showing that MSTs often organize assets according to sectoral and geographical structure. Song et al. [36] studied the evolution of worldwide stock markets through correlation-based graphs, documenting time variation in network topology.

The Planar Maximally Filtered Graph (PMFG) relaxes the sparsity of the MST while preserving planarity [40]. For N nodes, a PMFG contains $3N - 6$ edges and includes the MST as a subgraph when both are constructed from the same ranked distance matrix. Because the PMFG can retain triangles and 4-cliques, it is more suitable for studying local connectivity, clustering, hub reorganization and mesoscopic structure. Tumminello et al. [41] further explored the relationship between correlation, hierarchies and networks in financial markets. Massara et al. [26] later introduced the Triangulated Maximally Filtered Graph (TMFG), which extends this class of planar filtering methods and improves scalability for large systems.

Filtered networks are particularly relevant for risk analysis. Pozzi et al. [32] used PMFG-based analysis to show that peripheral assets in financial networks can offer better diversification properties than central hubs. Aste et al. [2] studied correlation dynamics in volatile markets using filtered network approaches. Kenett et al. [19] used partial correlation analysis to reveal the dominant role of the financial sector in stock market networks. These studies motivate the hub-reorganization analysis presented here. By comparing original and RMT-filtered networks, this paper asks whether centrality is driven mainly by market-wide co-movement or whether localized central nodes emerge after removing the dominant market mode.

2.4. Financial networks in emerging markets

Network methods have also been applied to emerging markets, where market structure can differ substantially from that of large developed equity markets. Emerging markets may combine concentrated sectoral composition, macroeconomic sensitivity, lower liquidity in parts of the market and exposure to global risk cycles. These features make them attractive systems for dependency analysis because the correlation structure may reflect both domestic and international channels.

Tabak et al. [39] studied the topology of the Brazilian interbank network, documenting structural properties and vulnerability to targeted failures. Silva et al. [34] analysed the structure and dynamics of the global financial network, with attention to how emerging economies connect to the broader system. Wang et al. [42] studied the correlation structure of world stock markets using Pearson and partial-correlation networks, showing that emerging markets can display distinctive dependency patterns.

Studies focused more directly on Brazilian markets include Pereira et al. [29], who applied detrended cross-correlation analysis to construct multiscale networks of stock markets including B3, and Leonel Caetano and Yoneyama [22], who applied complex network analysis to the Brazilian stock market. Stosic et al. [38] studied correlations and cross-correlations in Brazilian agrarian commodities and stocks. These studies show that Brazilian financial data contain rich structural properties, but the literature remains fragmented across interbank networks, commodities, cross-market relations and equity-market topology.

The present paper contributes to this literature by focusing on a broad B3 equity universe and by connecting spectral filtering, planar networks and volatility forecasting in a single empirical design. This integration matters because the interpretation of a financial network depends on the matrix from which it is built. A network constructed from the original correlation matrix may be dominated by the market mode, while a network constructed from group-mode components may reveal weaker but more localized dependency channels. This distinction is especially relevant in emerging markets, where common macroeconomic exposure and sectoral heterogeneity coexist.

2.5. Volatility forecasting and machine learning

Volatility forecasting has a long econometric tradition. ARCH and GARCH-type models, introduced by Engle [13] and generalized by Bollerslev [4], model conditional heteroskedasticity through parsimonious dynamic equations. Hansen and Lunde [16] showed that the simple GARCH(1,1) is difficult to outperform systematically. HAR-RV models [12] approximate long-memory behaviour in realized volatility through daily, weekly and monthly components. The theoretical foundations of realized-volatility measurement were established by Andersen et al. [1]. Poon and Granger [31] provide an influential review of volatility forecasting methods.

Forecast evaluation is delicate because realized volatility is an observable proxy for latent conditional variance. Loss functions can rank models differently depending on their robustness to measurement error and their treatment of volatility underestimation. QLIKE is widely used because of its robustness properties under noisy volatility proxies [28]. For this reason, the forecasting experiment in this paper evaluates models using multiple metrics, including MAE, RMSE, QLIKE and out-of-sample R^2.

Machine-learning methods provide flexible nonlinear approximations and can incorporate large feature sets. Random Forests [6], gradient boosting [14] and regularized linear models [17] can combine lagged volatility, rolling statistics, market-wide variables and network centrality measures. Bucci [7] applied neural networks to realized-volatility forecasting, showing that flexible models can capture nonlinear volatility dynamics. Christensen et al. [10] developed a machine-learning framework for volatility forecasting and emphasized the importance of careful feature engineering. Zhang and Broadstock [43] explored how financial network topology relates to market risk from a machine-learning perspective.

In financial forecasting, however, flexibility can amplify noise and produce unstable gains if temporal validation is not strict. For this reason, this paper evaluates machine-learning models against strong econometric baselines and treats network features as potential incremental predictors rather than replacements for classical volatility dynamics. The nested feature-set design follows an ablation logic: Set A measures the predictive value of classical volatility information, Set B tests the incremental value of market and RMT variables, and Set C tests whether MST and PMFG topology add further information beyond these baseline predictors.

3. Data and asset universe

3.1. BovDB/B3 data source

The empirical analysis uses Brazilian equity data derived from BovDB and its extended version BovDBv2. BovDB was originally proposed as an open dataset of stock prices for companies traded on the Brazilian Stock Exchange (B3), covering the period from 1995 to 2020 and applying cumulative adjustment factors to correct prices for corporate events [9]. BovDBv2 is distributed through Harvard Dataverse as the *Dataset for Intraday Analysis of B3 stock prices* [37], containing daily price information from 1995 to 2024 and intraday price data for Brazilian stocks from January 1995 to July 2024.

Although BovDBv2 includes intraday information, the present paper uses daily adjusted prices because the objective is to study long-run cross-sectional dependencies, Random Matrix Theory filtering and network topology over a broad historical window. Daily data provide a stable sampling frequency for estimating correlations across many assets and reduce microstructure effects that would be more prominent in intraday analysis. The main empirical objects are adjusted closing prices, logarithmic returns and correlation matrices constructed from synchronized daily observations.

3.2. Adjusted prices

The empirical analysis uses adjusted closing prices because long historical equity samples are affected by stock splits, reverse splits, dividends, interest on equity, bonuses, corporate reorganizations and share-class changes. Without adjustment, these events can generate artificial jumps that contaminate return distributions, correlations and network topology. Return construction, standardization and correlation estimation are defined formally in Section 4.

3.3. Asset filters and equity universe definition

The core equity universe is obtained through deterministic filters applied to the BovDB/B3 daily database. The starting point is the set of B3 instruments with daily adjusted price information. The analysis then excludes instruments that are not direct listed equities, including Brazilian Depositary Receipts (BDRs), exchange-traded funds (ETFs), investment funds, benchmark indices and other non-equity vehicles. The purpose of this step is to keep the dependency matrix focused on listed equity shares rather than mixing equities with instruments that mechanically track baskets, benchmarks or foreign securities.

After this class filter, observations are retained only when adjusted closing prices are valid and strictly positive. Daily logarithmic returns are computed from adjusted closing prices, and extreme adjusted-return observations are inspected as part of the data-quality procedure. These observations are used as exclusion criteria only when they indicate clear corporate-action adjustment inconsistencies. Assets are then required to satisfy a minimum historical-coverage condition, and the remaining assets are synchronized on common trading dates. The formal notation for this synchronized return panel is given in Section 4.1.

3.4. Core historical universe

The core historical universe contains 58 assets with sufficient data coverage for the main RMT, clustering and network analysis. This universe is intentionally historical rather than purely current. It allows the paper to study a long Brazilian equity sample while keeping the cross-section large enough for network analysis and small enough for interpretable visualizations.

The main spectral analysis uses a complete synchronized panel with $N = 58$ assets and $T = 1527$ observations. The complete-case restriction is important because the empirical correlation matrix must be estimated from a common set of trading dates across all assets. This avoids pairwise changes in sample composition and ensures that eigenvalues, eigenvectors, distances and network edges are derived from a single consistent correlation matrix.

3.5. Demonstration and forecasting assets

PETR4, VALE3 and BBDC4 are used as representative assets for stylized facts and as the initial forecasting targets. They were selected because they are large, liquid and economically distinct Brazilian equities. PETR4 represents oil and gas exposure, with sensitivity to commodity prices and firm-specific institutional events. VALE3 represents basic materials and global mining exposure, with sensitivity to commodity cycles and exchange-rate dynamics. BBDC4 represents the financial sector and domestic credit conditions.

This small panel is not intended to represent the entire Brazilian equity market. It provides a compact and interpretable entry point into the empirical analysis, illustrating heavy tails, volatility clustering, return extremes, absolute-return persistence and the feasibility of the forecasting design.

3.6. Sector and subsector classification

Each asset in the core historical universe is assigned to a macro-sector, subsector and segment classification. The macro sectors include Financials, Oil Gas and Biofuels, Basic Materials, Utilities, Industrials, Consumer Cyclical, Consumer Non-Cyclical, Real Estate and Telecommunications. These labels are used for descriptive comparison, sectoral correlation tests, heatmap interpretation and subsector-level aggregation.

Utilities and Real Estate are kept separate because their risk drivers differ substantially. Utilities are more directly linked to regulation, concessions, inflation adjustment mechanisms and defensive cash flows. Real Estate is more sensitive to credit conditions, interest rates and the real-estate cycle. Within Basic Materials, the subsector classification distinguishes Mining, Steel and Metallurgy, Chemicals, and Paper and Cellulose. This distinction is relevant in Brazil because commodity-linked firms can share global exposure while still belonging to different industrial chains.

Table 1: Summary of the empirical samples used in this study.

Sample	Assets	Period	Main use
Representative assets	3	2006–2025	Stylized facts and initial forecasting experiment
Core historical universe	58	1998–2025	Correlation, RMT, clustering and networks

3.7. Data quality notes and limitations

The main data-quality issue is the distinction between genuine extreme returns and artificial jumps caused by corporate-action adjustment problems. Adjusted prices reduce the impact of splits, dividends and related events, but they do not guarantee that every historical corporate action is perfectly corrected. This limitation is especially relevant in long emerging-market samples, where ticker histories, share-class changes and corporate reorganizations can be complex.

For PETR4, VALE3 and BBDC4, the largest daily returns observed in the 2006–2025 window are consistent with stress periods such as the 2008 global financial crisis, the Brazilian recession period around 2015–2016 and the 2020 COVID shock. Extreme-return validation is treated as a separate quality-control step, and assets with implausible corporate-action-related jumps are excluded from the main analyses.

The main empirical claims of the paper are based on the core historical universe and on synchronized daily returns. This choice provides a consistent synchronized panel for estimating correlation matrices, eigenvalue spectra and network topology, but it also imposes limits. Results may differ under intraday sampling, alternative liquidity filters, rolling-window universes, or broader target sets for forecasting. These extensions are left for robustness analysis and future work.

4. Methodology

The empirical design follows an integrated methodology that transforms adjusted B3 prices into return matrices, correlation matrices, spectral components, filtered financial networks and volatility-forecasting features. Figure 1 summarizes the workflow. The purpose is not only to estimate dependencies, but to distinguish market-wide motion, group-level structure and noise-compatible components before evaluating whether the resulting topology contains predictive information.

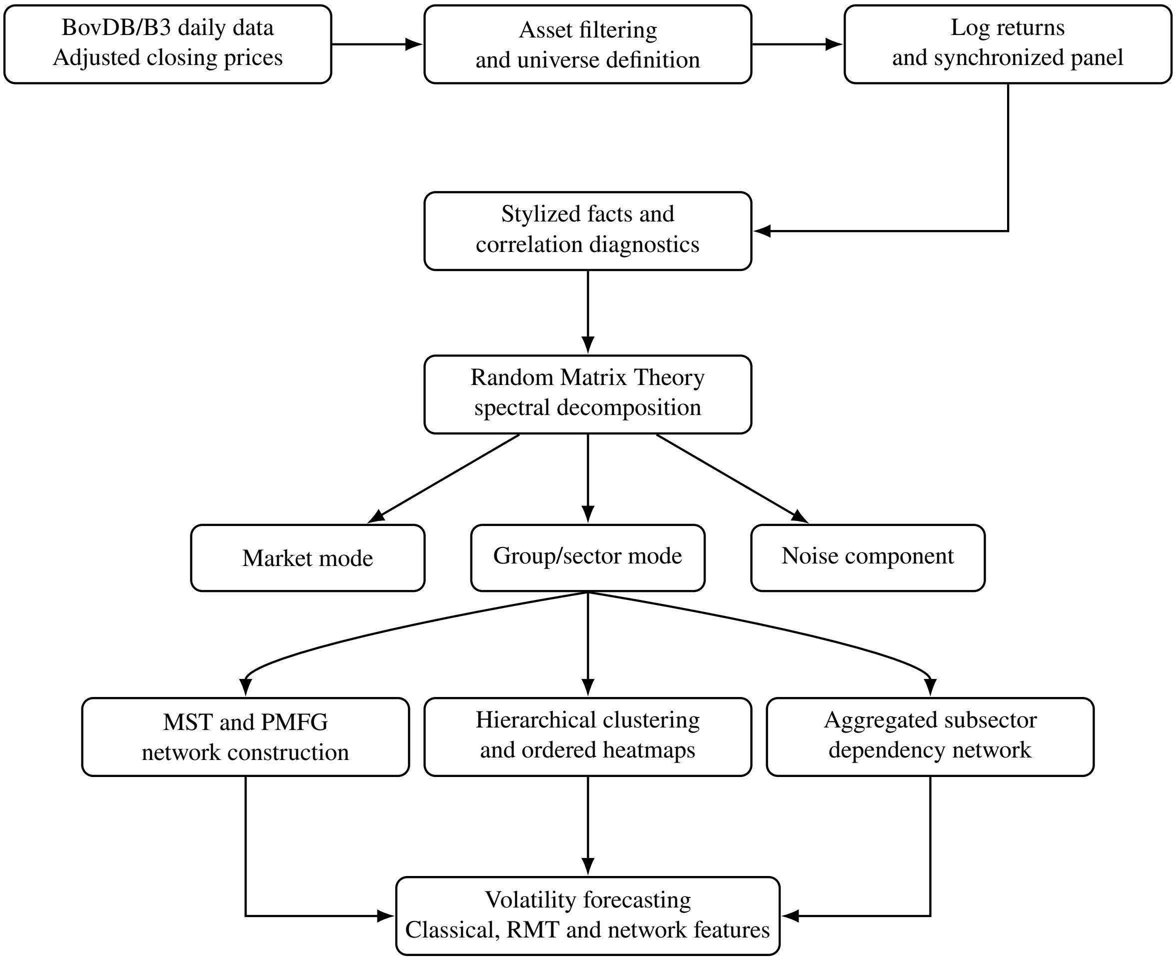


Figure 1: Overview of the empirical pipeline. Adjusted daily B3 prices are filtered to define the equity universe and construct synchronized return panels. The analysis then proceeds through stylized-fact diagnostics, correlation estimation, Random Matrix Theory filtering, hierarchical clustering, MST/PMFG network construction, subsector aggregation and volatility forecasting with classical, RMT and network-based features.

4.1. Notation and empirical return matrix

The notation is fixed before the empirical steps are defined. The index $i = 1, \dots, N$ identifies assets, and the index $t = 1, \dots, T$ identifies trading days in a synchronized panel. In the main spectral, clustering and network analysis, $N = 58$ is the number of assets and $T = 1527$ is the number of common daily observations after synchronization. Thus T is the number of dates in the complete panel, not the total number of scalar return entries.

Let $P_{i,t}^{\mathrm{adj}}$ denote the adjusted closing price of asset i on trading day t. Daily logarithmic returns are computed as

$$r_{i,t} = \log P_{i,t}^{\mathrm{adj}} - \log P_{i,t-1}^{\mathrm{adj}}, \tag{1}$$

where $P_{i,t}^{\mathrm{adj}} > 0$ and $P_{i,t-1}^{\mathrm{adj}} > 0$. Log returns are used because they are additive across time and commonly used in empirical studies of financial fluctuations.

The return panel is represented by the matrix

$$\mathbf{R} = \left(r_{i,t}\right)_{t=1,\dots,T;\, i=1,\dots,N} \in \mathbb{R}^{T\times N}. \tag{2}$$

Rows correspond to synchronized trading days and columns correspond to assets. For correlation-based analysis, each return series is standardized within the relevant sample:

$$\tilde{r}_{i,t} = \frac{r_{i,t} - \mu_i}{\sigma_i}, \tag{3}$$

where μ_i and σ_i are the sample mean and standard deviation of asset i. The standardized return matrix is denoted by $\widetilde{\mathbf{R}}$.

The empirical correlation matrix is denoted by $\mathbf{C}$, and C_{ij} is the Pearson correlation between assets i and j. In the Random Matrix Theory step, λ_k denotes the k-th eigenvalue of $\mathbf{C}$, $\mathbf{u}_k$ denotes its normalized eigenvector and $Q = T/N$ denotes the sample-size-to-dimension ratio used in the Marčenko–Pastur benchmark.

4.2. Asset filtering and synchronized panels

Let $\mathcal{A}_0$ denote the initial set of B3 instruments available in the BovDB/B3 daily database. The data section explains the economic motivation for excluding non-direct-equity instruments. Formally, let $\mathcal{X}$ be the excluded set containing BDRs, ETFs, investment funds, benchmark indices and other non-equity vehicles. The eligible direct-equity set is

$$\mathcal{A}_1 = \mathcal{A}_0 \setminus \mathcal{X}, \tag{4}$$

where each element of $\mathcal{A}_1$ is an asset identifier rather than an observation date.

For each remaining asset, observations are retained only when adjusted closing prices are valid:

$$P_{i,t}^{\mathrm{adj}} > 0. \tag{5}$$

Daily log returns are then computed from adjusted prices. Assets with insufficient historical coverage are removed according to

$$T_i \geq T_{\min}, \tag{6}$$

where T_i is the number of valid return observations for asset i and $T_{\min}$ is the minimum coverage threshold required for the target analysis.

After filtering, assets are synchronized by common trading dates. The RMT, clustering and network analyses are performed on a complete return panel

$$\mathbf{R}_{\mathrm{core}} \in \mathbb{R}^{T\times N}, \tag{7}$$

with $N = 58$ assets and $T = 1527$ complete daily observations in the main spectral analysis. This complete-panel restriction ensures that all eigenvalues, eigenvectors, distances and network edges are derived from a single consistent correlation matrix estimated from assets observed on common trading dates.

4.3. Stylized facts and descriptive diagnostics

The first empirical stage documents stylized facts for representative B3 assets. For each asset, the analysis computes the sample mean, standard deviation, skewness, excess kurtosis, minimum and maximum returns, empirical value-at-risk at 1% and 5%, counts of large absolute returns and autocorrelations of absolute returns.

For an asset return series $\{r_t\}$, the empirical autocorrelation of absolute returns at lag ℓ is

$$\rho_{|r|}(\ell) = \mathrm{Corr}(|r_t|, |r_{t-\ell}|). \tag{8}$$

These diagnostics assess whether the selected Brazilian return series display empirical regularities commonly associated with financial time series: heavy tails, volatility clustering and persistent volatility [11].

4.4. Correlation matrix estimation

The empirical Pearson correlation matrix is estimated from standardized returns as

$$C_{ij} = \frac{1}{T-1} \sum_{t=1}^{T} \tilde{r}_{i,t}\tilde{r}_{j,t}. \tag{9}$$

Equivalently, in matrix form,

$$\mathbf{C} = \frac{1}{T-1} \widetilde{\mathbf{R}}^{\top}\widetilde{\mathbf{R}}. \tag{10}$$

The matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$ is symmetric and has unit diagonal. For pairwise descriptive analysis, correlations may be computed with available observations for each pair. For RMT, clustering, MST and PMFG construction, the complete synchronized panel is used to obtain a single consistent $N \times N$ matrix.

4.5. Sectoral correlation comparison

Each asset is assigned to a macro-sector and subsector. For each pair (i, j), define

$$s_{ij} = \begin{cases} 1, & \text{if assets } i \text{ and } j \text{ belong to the same macro-sector,} \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Pairs with $s_{ij} = 1$ form the within-sector distribution, while pairs with $s_{ij} = 0$ form the between-sector distribution. The two empirical distributions are compared using mean, median, dispersion and a one-sided Mann–Whitney U test. This step evaluates whether the economic sector classification is reflected in the empirical dependency matrix before any RMT filtering is applied.

4.6. Rolling market synchronization

Static full-sample correlations hide time variation in market dependence. To measure market synchronization, rolling pairwise correlations are estimated over a window of length w. For each window ending at time t, the average market correlation is

$$\bar{\rho}_t^{(w)} = \frac{2}{N(N-1)} \sum_{1\le i<j\le N} \rho_{ij,t}^{(w)}, \tag{12}$$

where $\rho_{ij,t}^{(w)}$ is the Pearson correlation between assets i and j computed within the rolling window. Elevated values of $\bar{\rho}_t^{(w)}$ indicate stronger market-wide synchronization and motivate the subsequent spectral decomposition.

4.7. Random Matrix Theory filtering

Random Matrix Theory is used to distinguish informative collective modes from eigenvalue fluctuations compatible with sampling noise. Let the spectral decomposition of the empirical correlation matrix be

$$\mathbf{C} = \sum_{k=1}^{N} \lambda_k \mathbf{u}_k \mathbf{u}_k^{\top}, \tag{13}$$

where λ_k is the k-th eigenvalue and $\mathbf{u}_k$ is the corresponding normalized eigenvector. The term $\mathbf{u}_k\mathbf{u}_k^\top$ is the outer product that reconstructs the matrix contribution of mode k. Eigenvalues are sorted in decreasing order, so $k = 1$ denotes the largest empirical mode.

For a random correlation matrix estimated from N independent standardized time series of length T, the Marčenko–Pastur bounds are

$$\lambda_\pm = \sigma^2 \left(1 + \frac{1}{Q} \pm 2\sqrt{\frac{1}{Q}}\right), \qquad Q = \frac{T}{N}. \tag{14}$$

Since standardized returns are used, $\sigma^2 = 1$. Eigenvalues inside $[\lambda_-, \lambda_+]$ are interpreted as compatible with random-matrix noise, while eigenvalues above λ_+ are treated as candidates for informative collective modes.

Let K denote the number of empirical eigenvalues above λ_+. The dominant market component is defined as

$$\mathbf{C}^{\text{market}} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top. \tag{15}$$

The group or sector component is reconstructed from the remaining informative eigenvalues:

$$\mathbf{C}^{\text{group}} = \sum_{k=2}^{K} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top. \tag{16}$$

The filtered informative matrix is

$$\mathbf{C}^{\text{filtered}} = \sum_{k=1}^{K} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top, \tag{17}$$

and the residual noisy component is

$$\mathbf{C}^{\text{noise}} = \sum_{k=K+1}^{N} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top. \tag{18}$$

Note that partial spectral reconstructions generally do not preserve the unit diagonal of the original correlation matrix. When these filtered matrices are used for downstream tasks, they are rescaled to unit diagonal if the subsequent method requires valid Pearson correlation bounds. This decomposition allows the analysis to compare networks dominated by broad market co-movement with networks that emphasize localized group and sectoral structure.

4.8. Mantegna distance and hierarchical clustering

Correlations are converted into distances using the metric introduced by Mantegna [23]:

$$d_{ij} = \sqrt{2(1 - C_{ij})}. \tag{19}$$

For a valid correlation matrix, this transformation maps strongly positively correlated assets to short distances and less positively correlated or negatively correlated assets to longer distances. The resulting distance matrix is used for hierarchical clustering, ordered heatmaps, MST and PMFG construction.

Hierarchical clustering is applied to the distance matrix using average linkage (UPGMA). The cophenetic correlation [35] is used to assess how faithfully the dendrogram preserves pairwise distances. Ordered heatmaps use the dendrogram ordering to rearrange the correlation matrices, making block structures visually identifiable. Comparing original, filtered and group-mode heatmaps helps distinguish broad market co-movement from localized sectoral dependency.

4.9. Minimum Spanning Tree and Planar Maximally Filtered Graph

A financial network is represented as

$$G = (V, E, w), \tag{20}$$

where V is the set of assets, E is the retained set of edges and $w_{ij} = C_{ij}$ is the dependency weight associated with edge (i, j).

The Minimum Spanning Tree is the connected acyclic graph that minimizes total distance:

$$\mathrm{MST} = \arg\min_{E'} \sum_{(i,j)\in E'} d_{ij}, \tag{21}$$

subject to (V, E') being connected and acyclic. For N nodes, the MST has $N-1$ edges. With $N = 58$, this gives 57 edges.

The Planar Maximally Filtered Graph is constructed by considering candidate edges in increasing order of distance, retaining them if they do not violate planarity. For N nodes, a PMFG has

$$|E_{\mathrm{PMFG}}| = 3N - 6 \tag{22}$$

edges, which gives 168 edges for $N = 58$. Unlike the MST, the PMFG can contain cycles, triangles and 4-cliques. Under the same ranking of pairwise distances, the MST is contained in the PMFG as a subgraph. This makes the PMFG a less aggressive topological filter that preserves richer local geometry.

For each network, the analysis reports edge correlations, same-sector and same-subsector edge ratios, average clustering, node degree, betweenness centrality and hub rankings. Comparing original and group-mode networks identifies whether centrality is driven by broad market co-movement or by localized dependency channels.

4.10. Aggregated subsector dependency network

The asset-level networks are complemented by an aggregated subsector dependency network. Let $g(i)$ denote the subsector associated with asset i. For two subsectors a and b, the aggregated dependency weight is defined as

$$W_{ab} = \frac{1}{|\mathcal{E}_{ab}|} \sum_{(i,j)\in\mathcal{E}_{ab}} w_{ij}, \tag{23}$$

where

$$\mathcal{E}_{ab} = \{(i,j) \in E \,:\, g(i) = a,\ g(j) = b\}. \tag{24}$$

This aggregation converts ticker-level dependencies into a mesoscopic map of subsector-level dependency channels. The resulting network is summarized using edge density, average dependency, weighted degree and betweenness centrality.

4.11. Volatility forecasting formulation

The forecasting task is formulated as a supervised time-series prediction problem. For target asset i and horizon h, realized volatility is defined as

$$RV_{i,t,h} = \sqrt{\sum_{\tau=1}^{h} r_{i,t+\tau}^2}. \tag{25}$$

The objective is to predict $RV_{i,t,h}$ using an information set $\mathcal{F}_t$ available at time t:

$$\widehat{RV}_{i,t,h} = f_\theta(\mathcal{F}_t). \tag{26}$$

For forecast evaluation under metrics such as QLIKE, these realized volatility and predicted volatility measures are squared to represent realized variance proxies. The target horizons are $h = 5$ and $h = 20$ trading days. The initial forecasting experiment uses PETR4, VALE3 and BBDC4, with a strict temporal split into training, validation and test periods to reduce look-ahead bias and preserve the chronological ordering of the forecasting experiment.

4.12. Econometric benchmarks and machine-learning feature sets

The benchmark set includes historical volatility, EWMA, GARCH(1,1) and HAR-RV. The GARCH(1,1) conditional variance is

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{27}$$

The GARCH models are estimated with Student-t innovations to allow for heavy-tailed return innovations, consistent with the stylized-fact diagnostics. HAR-RV uses daily, weekly and monthly realized-volatility components to approximate long-memory behaviour [12]. These benchmarks are intentionally strong because the GARCH(1,1) specification is notoriously difficult to outperform systematically [16].

Three feature sets are used in a nested ablation design. Set A contains classical return and volatility predictors, including rolling volatilities, rolling absolute returns, lagged realized volatility, EWMA volatility and higher-moment rolling statistics. Set B adds market and RMT variables such as the largest eigenvalue, market-mode trace share and number of eigenvalues above the random-matrix bound. Set C adds MST and PMFG features, including degree, betweenness and clustering measures from original and group-mode networks.

This nesting defines a controlled ablation:

$$\mathcal{F}_t^A \subset \mathcal{F}_t^B \subset \mathcal{F}_t^C. \tag{28}$$

Set A measures the predictive value of classical volatility information alone. Set B tests the incremental value of market-wide and RMT features. Set C tests whether network topology adds further information beyond these baselines. Ridge regression, Random Forest and histogram gradient boosting are used to evaluate these increasingly rich information sets.

4.13. Forecast evaluation metrics

Forecasts are evaluated using MAE, RMSE, out-of-sample R^2 and QLIKE. For realized values y_t and forecasts $\hat{y}_t$, the MAE and RMSE are

$$\mathrm{MAE} = \frac{1}{M}\sum_{t=1}^{M} |y_t - \hat{y}_t|, \tag{29}$$

and

$$\mathrm{RMSE} = \sqrt{\frac{1}{M}\sum_{t=1}^{M}(y_t - \hat{y}_t)^2}. \tag{30}$$

The out-of-sample coefficient of determination is

$$R^2_{\mathrm{oos}} = 1 - \frac{\sum_{t=1}^{M}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{M}(y_t - \bar{y}_{\mathrm{train}})^2}. \tag{31}$$

The QLIKE loss is

$$\mathrm{QLIKE}_t = \log(\hat{\sigma}_t^2) + \frac{\sigma_t^2}{\hat{\sigma}_t^2}, \tag{32}$$

where $\hat{\sigma}_t^2$ is the predicted variance proxy and σ_t^2 is the realized variance proxy. QLIKE is included because it is widely used for volatility forecast comparison when realized volatility is an imperfect proxy for latent variance [28].

5. Stylized facts and correlation structure

5.1. Stylized facts of selected B3 assets

Figure 2 summarises the return properties of PETR4, VALE3 and BBDC4. These assets are used as representative examples and as the initial forecasting targets because they are liquid, economically important and sectorally distinct; they are not intended to be a proxy for the entire Brazilian equity market. The figure combines four diagnostics that are standard in the econophysics literature [11]: normalized prices, daily returns, the complementary cumulative distribution of absolute returns and the autocorrelation of absolute returns.

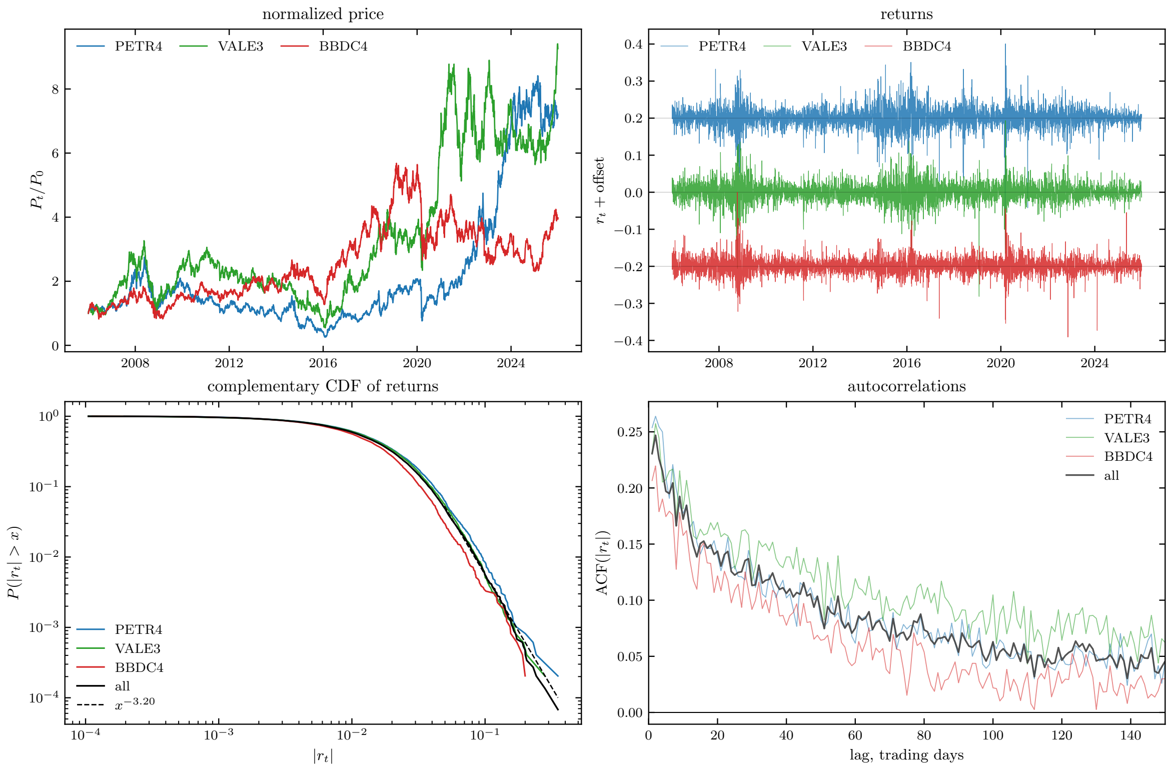


Figure 2: Stylized facts for PETR4, VALE3 and BBDC4 over 2006–2025. The panels show normalized prices, daily log returns, the complementary cumulative distribution of absolute returns and autocorrelations of absolute returns. In the return panel, the series are vertically offset only to improve visual separation; the offsets do not change the underlying daily log-return scale.

The normalized-price panel shows heterogeneous long-run trajectories across the three assets. The return panel shows volatility clustering, with large shocks concentrated around stress periods rather than uniformly distributed. The CCDF panel indicates that large absolute returns are more frequent than a Gaussian benchmark would predict. The right panel shows slowly decaying autocorrelation in absolute returns, consistent with persistent volatility dynamics [11].

5.2. Descriptive statistics

Table 2 quantifies the same diagnostics. All three series have means close to zero, substantial kurtosis and non-negligible counts of large absolute returns. PETR4 has the largest negative daily observation in the 2006–2025 window, while VALE3 and BBDC4 also show stress-period extremes. The autocorrelations of absolute returns remain positive across several lags, especially for VALE3.

The descriptive statistics are consistent with the visual reading of Figure 2. The return distributions display substantial departures from a Gaussian benchmark, including excess kurtosis and persistent absolute-return autocorrelations. These properties are consistent with the well-documented stylized facts of financial returns [11] and support treating the Brazilian market as a complex empirical system in which dependency structure, rather than marginal normality, is the relevant modelling object.

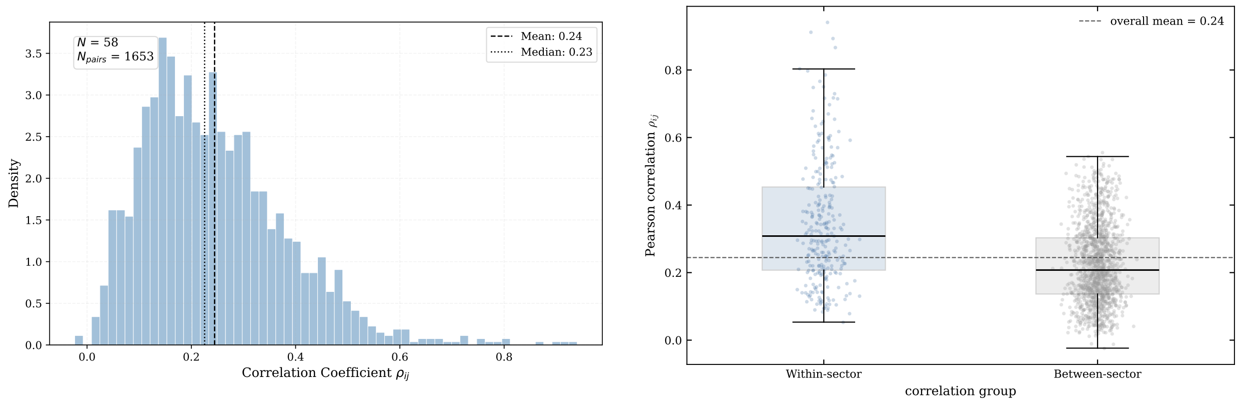
Table 2: Descriptive statistics for the three demonstration assets, 2006–2025.

Symbol	n	Mean	Std.	Skew.	Ex. kurt.	Min.	Max.	$\mathrm{VaR}_{1\%}$	$\mathrm{VaR}_{5\%}$	$\|r\| > 10\%$	$\|r\| > 20\%$	ACF_1	ACF_5	ACF_{20}	ACF_{60}
BBDC4	4953	0.00028	0.02180	0.02	7.96	-0.1910	0.1999	-0.0559	-0.0312	16	0	0.207	0.174	0.122	0.044
PETR4	4953	0.00041	0.02755	-0.76	11.55	-0.3524	0.2007	-0.0767	-0.0408	41	4	0.254	0.209	0.140	0.062
VALE3	4953	0.00045	0.02576	-0.22	7.51	-0.2814	0.1936	-0.0676	-0.0381	26	2	0.231	0.207	0.169	0.119

Notes: Mean and Std. are the sample mean and standard deviation of daily log returns. Skew. is sample skewness and Ex. kurt. is excess kurtosis relative to the Gaussian benchmark. $\mathrm{VaR}_{1\%}$ and $\mathrm{VaR}_{5\%}$ are the empirical 1% and 5% lower-tail return quantiles. $|r| > 10\%$ and $|r| > 20\%$ count daily observations whose absolute log return exceeds the stated threshold. ACF_ℓ is the autocorrelation of absolute returns at lag ℓ trading days.

5.3. Static correlation distribution

The next step moves from univariate behaviour to cross-sectional dependence. The primary object is the complete empirical correlation matrix for the 58-asset synchronized panel. This matrix is shown explicitly in the original-correlation panel of Figure 7; the distributional summaries below compress the same matrix into pairwise statistics before the RMT decomposition is discussed. The core historical universe contains 58 assets, producing $N(N-1)/2 = 1653$ unique asset pairs. Figure 3 summarizes the full pairwise correlation distribution and then splits pairs into within-sector and between-sector groups.



(a) Full pairwise distribution.

(b) Within-sector and between-sector pairs.

Figure 3: Correlation structure in the core historical universe. Panel (a) shows the full Pearson correlation distribution. Panel (b) compares within-sector and between-sector correlations using the initial macro-sector classification.

The full distribution is centred in a moderate positive range, with mean correlation close to 0.24 and median correlation close to 0.23. This level of average dependence suggests the presence of a common component, but the distribution remains dispersed. The dispersion is important because it leaves room for sectoral and subsectoral organization rather than a single homogeneous market block.

5.4. Sectoral correlation comparison

The sector split strengthens this interpretation. Within-sector correlations are visibly shifted to the right relative to between-sector correlations. This provides descriptive evidence that the initial macro-sector map is reflected in the empirical dependency matrix.

Table 3: Within-sector and between-sector correlation summary.

Group	Pairs	Mean	Median	Std. dev.	Min	Max
Within-sector	275	0.3433	0.3082	0.1819	0.0534	0.9402
Between-sector	1378	0.2249	0.2076	0.1166	-0.0236	0.5552

The difference is economically meaningful. Within-sector pairs have mean correlation 0.3433, while between-sector pairs have mean correlation 0.2249. The one-sided Mann–Whitney test for the alternative that within-sector correlations are larger gives $p = 1.33389 \times 10^{-24}$. Although the Mann–Whitney test assumes independent observations—an assumption violated by overlapping pairwise correlations—the large effect size and the matrix visualization in Figure 7 support the descriptive interpretation that sectoral groupings are reflected in the empirical dependency matrix.

5.5. Dynamic market synchronization

Full-sample correlations are limited because market dependence changes over time. Figure 4 shows the rolling average market correlation. This statistic compresses the correlation matrix into a time series that measures how synchronized the market is at each point in time.

The rolling average correlation rises during stress periods, including the 2008 global financial crisis, the Brazilian recession period around 2015–2016 and the 2020 COVID shock. These episodes indicate stronger market-wide synchronization. Such market-wide synchronization is consistent with the emergence of a dominant leading eigenvalue in the full-sample correlation matrix.

Figure 5 adds a more granular view by showing selected pairwise dynamic correlations under different estimators. The objective is not to replace the full matrix but to show that individual dependencies can change substantially over time.

The dynamic pairwise figure shows why a single full-sample correlation should be interpreted carefully. Short-window correlations react quickly but are noisy; longer-window estimates are smoother but slower; EWMA estimates provide an intermediate view. This descriptive evidence motivates the subsequent use of spectral filtering and networks as structured summaries of a noisy and time-varying dependency system.

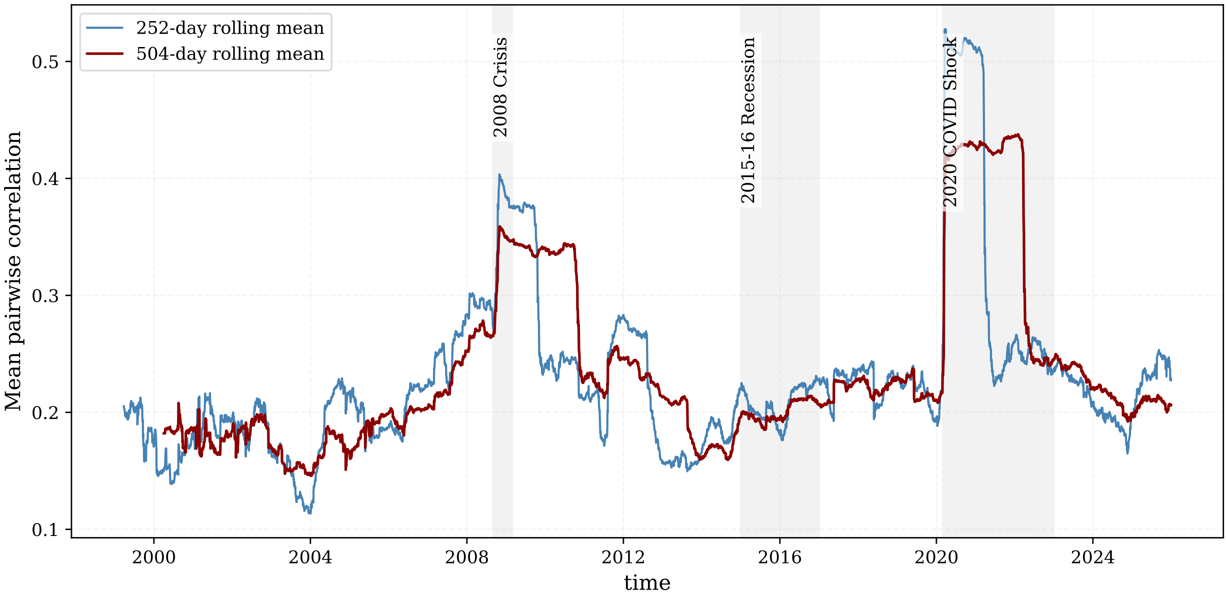


Figure 4: Rolling average market correlation for the core historical universe.

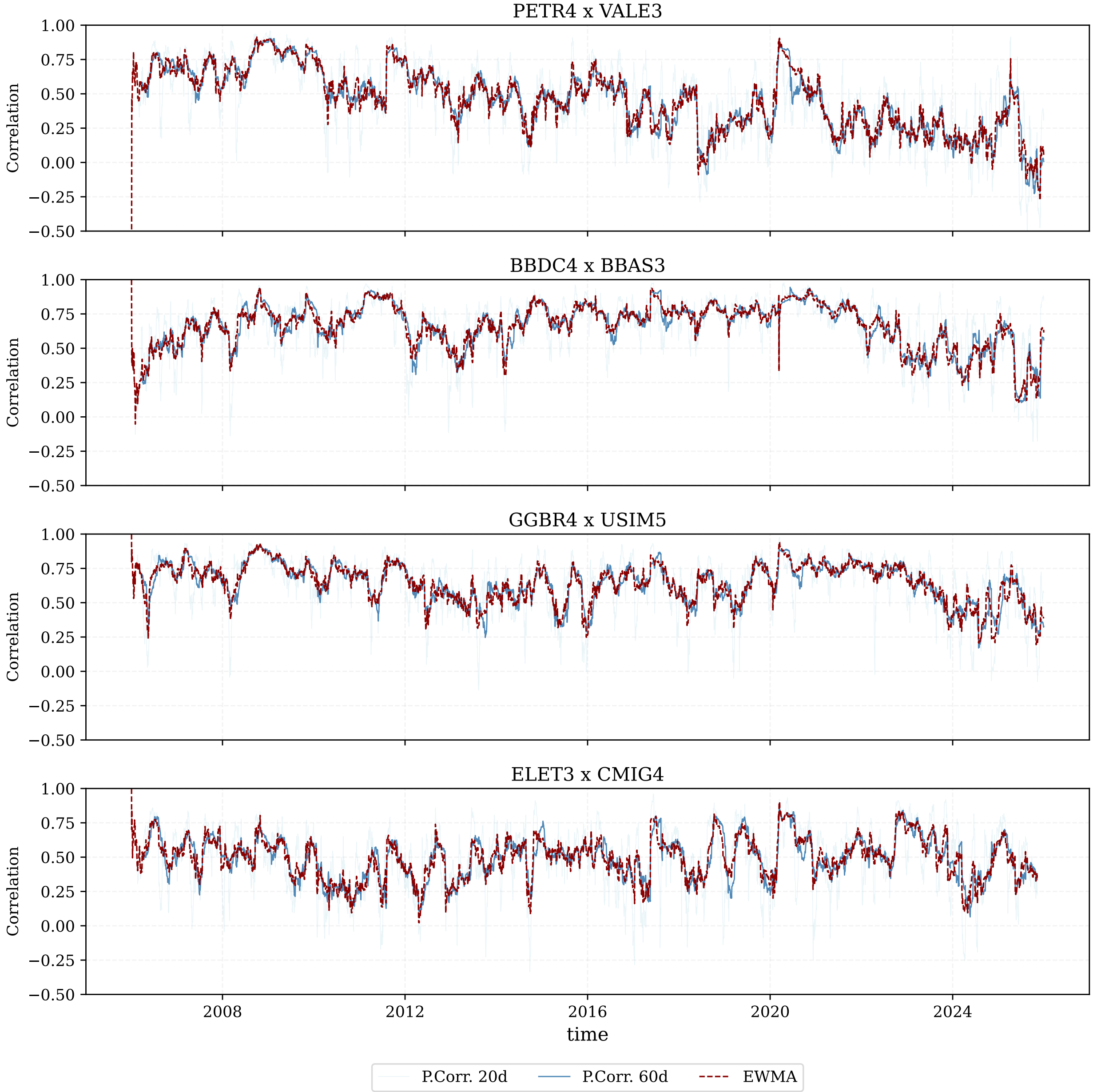


Figure 5: Dynamic pairwise correlations for selected asset pairs. Short-window, longer-window and EWMA estimates highlight the instability of individual dependencies.

6. Random Matrix filtering

6.1. Empirical eigenspectrum

Random Matrix Theory provides the spectral benchmark for distinguishing informative collective modes from sampling noise in the empirical correlation matrix [20, 30]. This separation measures the deviation of the empirical correlation matrix from the random-matrix benchmark. The core historical matrix uses $N = 58$ assets and $T = 1527$ observations, giving $Q = T/N \approx 26.33$. The Marčenko–Pastur bounds are $\lambda_- = 0.6482$ and $\lambda_+ = 1.4278$.

Table 4: RMT summary for the core historical universe.

N	T	Q	λ_-	λ_+	λ_1	λ_2	λ_3	λ_4	λ_5	Above λ_+
58	1527	26.33	0.6482	1.4278	21.6505	3.0957	1.7839	1.6543	1.4835	5

Table 4 provides the numerical basis for the RMT filtering step. With $N = 58$ assets and $T = 1527$ complete observations, the aspect ratio is $Q = 26.33$, which implies a relatively narrow Marčenko–Pastur noise band from $\lambda_- = 0.6482$ to $\lambda_+ = 1.4278$. The first eigenvalue, $\lambda_1 = 21.6505$, is therefore not a marginal deviation from the random-matrix benchmark but a dominant collective component. The next four eigenvalues, $\lambda_2 = 3.0957$, $\lambda_3 = 1.7839$, $\lambda_4 = 1.6543$ and $\lambda_5 = 1.4835$, also exceed the upper random bound. There are 5 empirical eigenvalues outside the band: these are the candidate collective modes, indicating one broad market mode and four additional modes that may be associated with group, sectoral or localized dependency structure.

Figure 6 shows the empirical spectrum against the random benchmark. The largest eigenvalue lies far above the Marčenko–Pastur upper bound. Four additional eigenvalues also exceed the upper edge, consistent with a dominant market mode and a small number of group or sector modes.

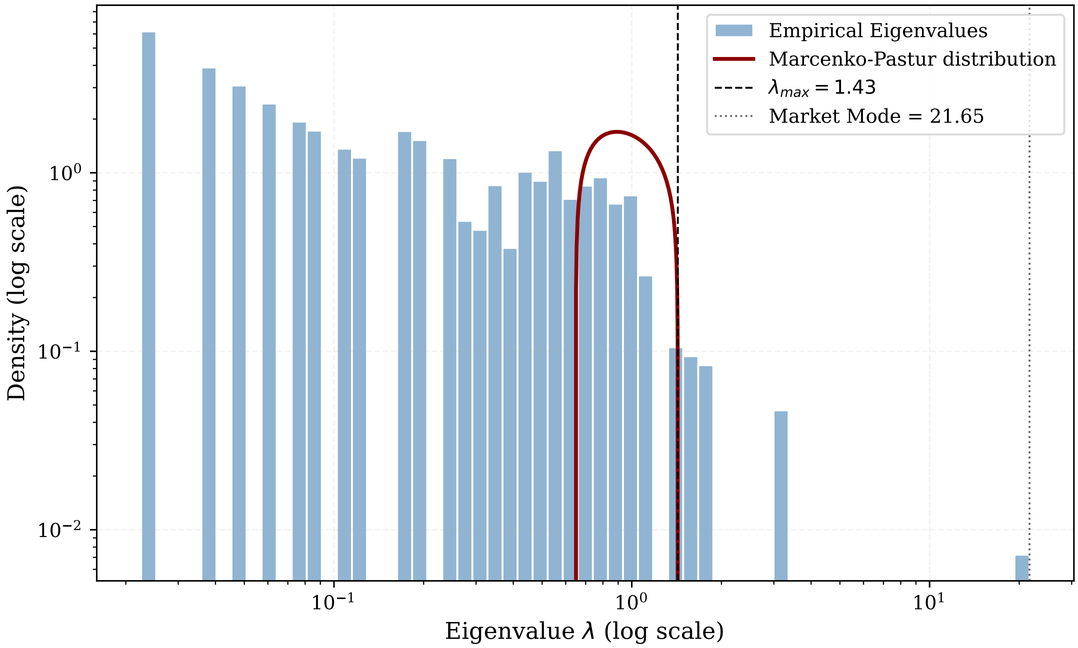


Figure 6: Empirical eigenvalue spectrum and Marčenko–Pastur bounds. The largest eigenvalue is far outside the random band and is interpreted as a market mode.

6.2. Market Mode and significant eigenvalues

The largest eigenvalue is 21.6505, corresponding to approximately 37.3% of the total trace of the correlation matrix. Because the associated eigenvector has positive loadings for nearly all assets and the eigenvalue lies far above the Marčenko–Pastur upper bound, this component is interpreted as a market mode. It captures broad co-movement and explains why the original correlation matrix has a strong positive common component. The next four eigenvalues are 3.0957, 1.7839, 1.6543 and 1.4835. These values are above λ_+ and are interpreted as candidates for group or sector structure.

The key methodological point is that the market mode is informative but also visually dominant. If a network is built directly from the original matrix, many edges can reflect market-wide co-movement rather than local structure. The group-mode matrix is therefore constructed to ask what remains after the broad common component is removed.

6.3. Matrix reconstruction

The reconstruction step checks that the spectral decomposition and the filtered matrices are being computed consistently. If all eigenvalue–eigenvector modes are recombined, the original correlation matrix is recovered up to floating-point precision. Table 5 reports this numerical check: the Frobenius reconstruction error is approximately 2.74×10^{-14}, and the maximum absolute reconstruction error is approximately 3.99×10^{-15}. The reported unit diagonal refers to the filtered matrix after the partial reconstruction has been rescaled for downstream correlation-based distances. This check reduces the possibility that subsequent network differences are driven by numerical reconstruction error rather than by the intended selection of market, group and noise-compatible modes.

Table 5: RMT reconstruction diagnostics.

Diagnostic	Value
Frobenius reconstruction error	2.74×10^{-14}
Maximum absolute reconstruction error	3.99×10^{-15}
C_{filtered} diagonal min	1.0000
C_{filtered} diagonal max	1.0000
Eigenvalues above λ_+	5
Market-mode trace share	0.3733

Figure 7 shows the resulting matrix components. The original-correlation panel is the complete empirical correlation matrix used throughout the paper. The market-mode matrix captures broad positive co-movement. The group-mode matrix displays weaker localized patterns after the market mode is removed. The noise component contains the random-band contribution, and the filtered matrix recombines the informative modes.

The relevant empirical point is that the eigenvectors beyond the leading component display loading patterns that are consistent with localized or sector-related dependencies. This residual organization motivates the clustering and network sections: if localized dependency patterns remain after market-mode removal, MST and PMFG topologies should differ when constructed from the group-mode matrix.

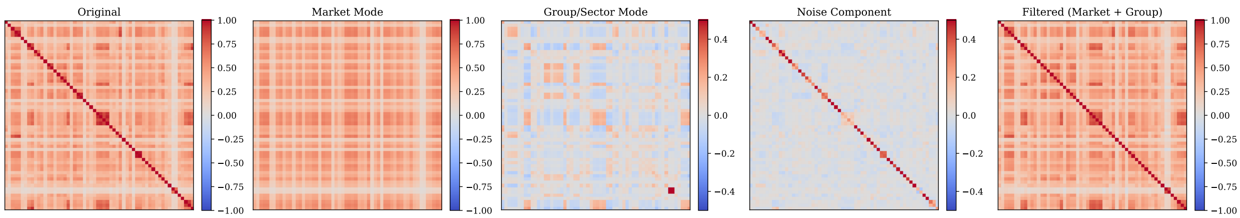


Figure 7: Empirical and RMT-filtered matrix components. The original-correlation panel shows the complete 58-asset correlation matrix; the remaining panels separate it into market, group/sector, noise and filtered components.

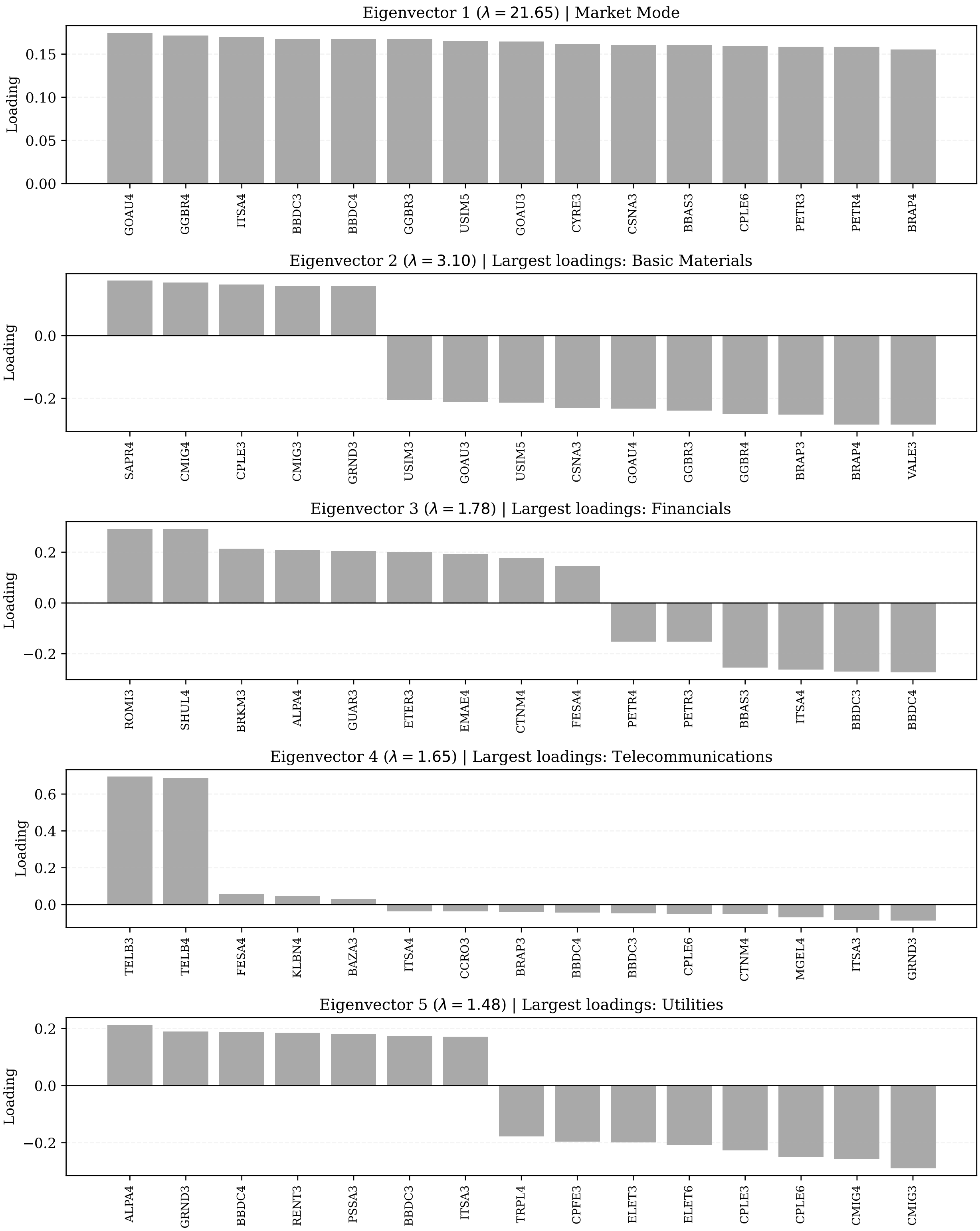


Figure 8: Top eigenvector loadings for the leading RMT modes. Eigenvector 1 is interpreted as the market-mode component; subsequent eigenvectors are interpreted cautiously as candidate group or localized sectoral components.

6.4. Top eigenvector loadings

Eigenvalues identify the strength of collective modes, but eigenvectors help interpret their composition. Figure 8 reports the largest loadings for the leading eigenvectors. The first eigenvector is interpreted as a market-mode component because its loadings are broadly positive across the cross-section. Subsequent eigenvectors are interpreted cautiously as candidate group or localized components.

For each eigenvector, assets are ranked by the magnitude of their loadings. The signs of eigenvector loadings are arbitrary up to multiplication by -1; therefore, interpretation focuses on relative grouping and magnitude rather than sign alone.

The second eigenvector is consistent with a Basic Materials and commodity-related component. The third contains financial and cross-sector structure. The fourth and fifth eigenvectors are more localized, with Telecommunications and Utilities-like components appearing among the dominant loadings. These labels are economic interpretations of the loading patterns, not mechanical classifications. Eigenvectors can mix sectors when firms share macroeconomic, liquidity or balance-sheet exposures.

7. Hierarchical clustering and ordered heatmaps

7.1. Ordered heatmaps

RMT decomposes the correlation matrix into spectral components, while ordered heatmaps provide a visual diagnostic of whether these components display block-like organization. Figure 9 orders the assets through hierarchical clustering and displays original, filtered and group-mode structures. The purpose is exploratory and diagnostic: the heatmaps assess whether spectral filtering is associated with patterns that can be interpreted in relation to sector and subsector classifications.

The original matrix displays a broad positive background, consistent with the large leading eigenvalue. The filtered matrix preserves part of this organized dependence while reducing the contribution of eigenvalues inside the random-matrix band. The group-mode matrix has lower average correlation by construction, but its ordered representation displays localized block-like patterns that are less dominated by the market-wide component.

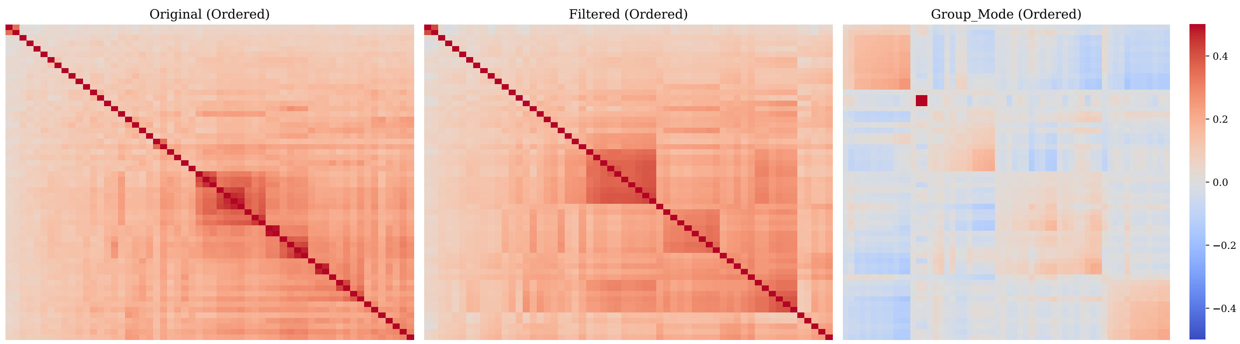


Figure 9: Ordered heatmaps for original and RMT-filtered correlation matrices. Hierarchical ordering reveals block structures that are less visible in unordered matrices.

7.2. Dendrograms comparison

Dendrograms provide a complementary view of the same distance geometry. Figure 10 compares hierarchical clustering trees for original, filtered and group-mode matrices. These trees help identify which assets merge at shorter distances and how the hierarchy changes after market-mode removal. Branch stability is not formally tested here; the dendrograms are used as exploratory summaries of the distance geometry.

Table 6: Cophenetic correlations for hierarchical clustering.

Matrix	Assets	Cophenetic correlation
Original	58	0.9500
Filtered	58	0.9340
Group Mode	58	0.8705

The cophenetic correlation is highest for the original matrix, 0.9500, and remains high for the filtered matrix, 0.9340. The group-mode value is lower, 0.8705. This decline is expected because removing the leading market component reduces the smooth common structure that is easier to summarize hierarchically. The lower value should therefore be interpreted as evidence that the group-mode distance geometry is less tree-like, not as evidence that it is uninformative.

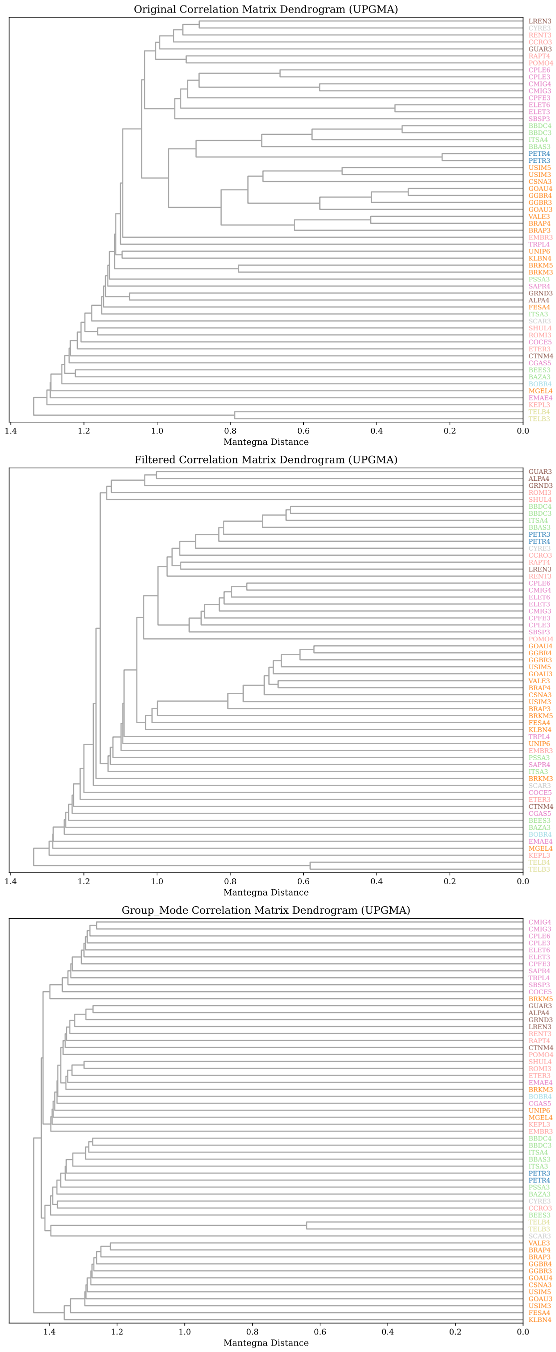


Figure 10: Dendrogram comparison for original, filtered and group-mode matrices using Mantegna distances and average linkage.

The clustering results connect naturally to network filtering. Ordered heatmaps and dendrograms suggest that localized structure remains after RMT filtering. The next section examines how this structure appears when only the strongest distance-ranked edges are retained under MST and PMFG constraints.

8. Financial network topology

Clustering summarises hierarchical organisation, but it does not identify graph-theoretic hubs, bridges or local cycles. Financial networks provide these complementary views by retaining selected high-correlation, short-distance edges under structural constraints. We therefore move from dendrograms to filtered financial networks to examine how dependency topology changes after RMT filtering.

8.1. MST backbone

The MST is the sparsest connected representation of the dependency geometry. With $N = 58$, it contains 57 edges. Figure 11 compares the MST built from the original matrix with the MST built from the group-mode matrix. This comparison illustrates how the MST backbone changes after the leading market component is removed.

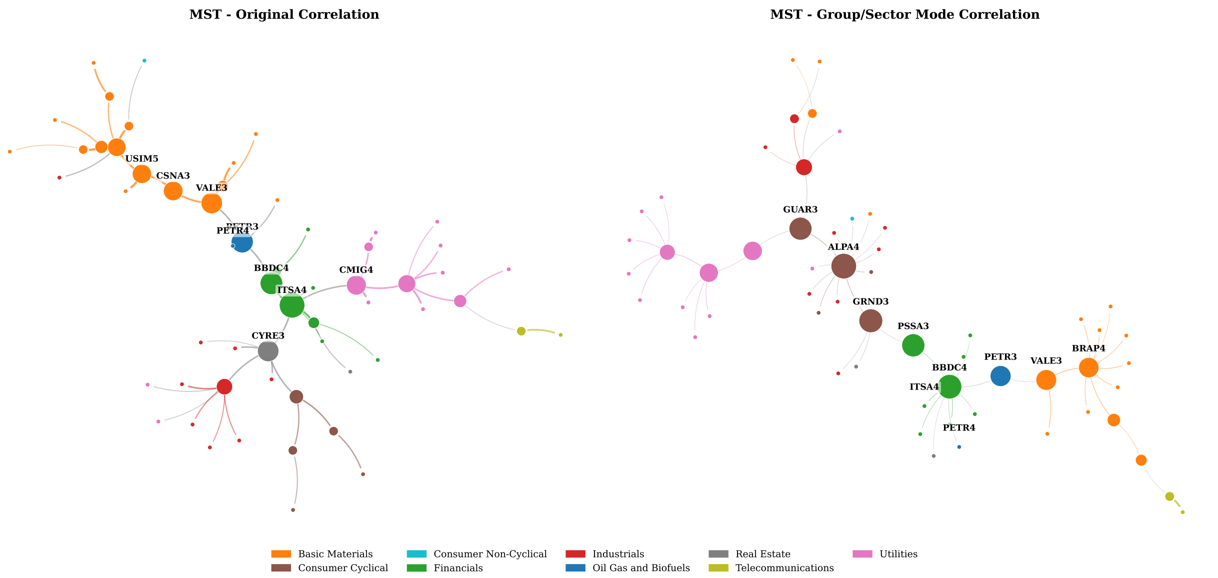


Figure 11: Refined MST comparison for original and group-mode correlation matrices. The original MST emphasizes strong market-mode dependencies, while the group-mode MST highlights weaker localized dependency channels.

The original MST has mean edge correlation 0.5796, while the group-mode MST has mean edge correlation 0.1388. This decline is expected because the group-mode network is constructed after excluding the dominant common component. The same-sector edge ratio also changes, from 0.7193 in the original MST to 0.6316 in the group-mode MST. The group-mode MST should therefore not be interpreted simply as a weaker version of the original tree. It represents a different dependency backbone constructed after removing the leading common component.

8.2. PMFG topology

The PMFG is less sparse than the MST because it preserves planarity while retaining $3N - 6 = 168$ edges. It can therefore retain cycles, triangles and 4-cliques, which are absent from the MST. Figure 12 compares original and group-mode PMFGs. This is the main network visualization in the paper because the PMFG retains enough structure to study local connectivity patterns, clustering and hub reorganization.

The original PMFG has mean edge correlation 0.5177. The group-mode PMFG has mean edge correlation 0.1109. Despite this lower average edge strength, the group-mode PMFG has higher average clustering, 0.6653, than the original PMFG, 0.5273. This is an important result: removing the market mode lowers the absolute level of edge correlations, but the resulting PMFG is associated with stronger local clustering.

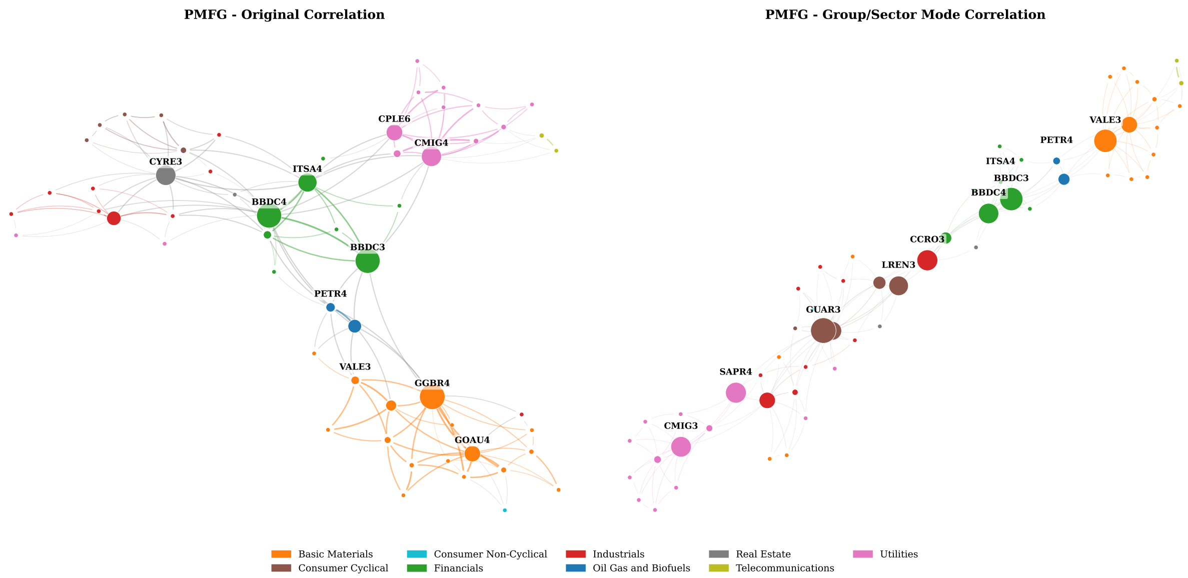


Figure 12: Refined PMFG comparison for original and group-mode correlation matrices. The group-mode PMFG reduces market-wide dominance and highlights more localized dependency channels.

8.3. Original vs Group/Sector Mode networks

Table 7 summarizes the main MST and PMFG statistics. Across both graph families, original networks display stronger average edge correlations, while group-mode networks display weaker edge correlations but retain nontrivial sectoral and topological structure. Original networks have stronger average edge correlations, as expected, because they are constructed from matrices that include the leading market component. Group-mode networks have weaker edges but retain patterns consistent with sectoral and subsectoral organization.

Table 7: Selected MST and PMFG topology statistics.

Network	Matrix	Edges	Mean edge corr.	Same-sector ratio	Clustering	Top degree	Top betweenness	4-cliques
MST	Original	57	0.5796	0.7193	0.0000	RAPT4	ITSA4	0
MST	Group mode	57	0.1388	0.6316	0.0000	ALPA4	ALPA4	0
PMFG	Original	168	0.5177	0.6012	0.5273	BBDC4	GGBR4	55
PMFG	Group mode	168	0.1109	0.5417	0.6653	GUAR3	GUAR3	55

8.4. Network topology metrics

Figure 13 expands the comparison beyond average edge correlation. It summarizes mean edge correlation, same-sector ratio, same-subsector ratio and average clustering across network and matrix types. This figure illustrates that topology should not be reduced to a single edge-weight statistic.

The topology comparison supports a mesoscopic interpretation. Group-mode networks have lower mean edge correlations, while the PMFG clustering coefficient increases. Same-sector and same-subsector ratios remain substantial after market-mode removal. This pattern is consistent with localized dependency structure, although a formal comparison with graph null models would be required to quantify how unlikely the topology is under randomized alternatives. MST clustering is zero by construction because trees do not contain triangles; the PMFG clustering coefficient is therefore the relevant statistic for local triangular organization.

Table 8: PMFG clique and planarity diagnostics.

Matrix	Nodes	Edges	Triangles	4-cliques	Planar
Original	58	168	166	55	Yes
Filtered	58	168	166	55	Yes
Group Mode	58	168	166	55	Yes

The clique counts are construction diagnostics. For a PMFG with 58 nodes, 168 edges are expected. The presence of triangles and 4-cliques shows that the PMFG retains local topological structures that are absent from the MST. These structures can be transformed into network predictors for the forecasting experiment, although their economic interpretation depends on the underlying edge weights and sector composition.

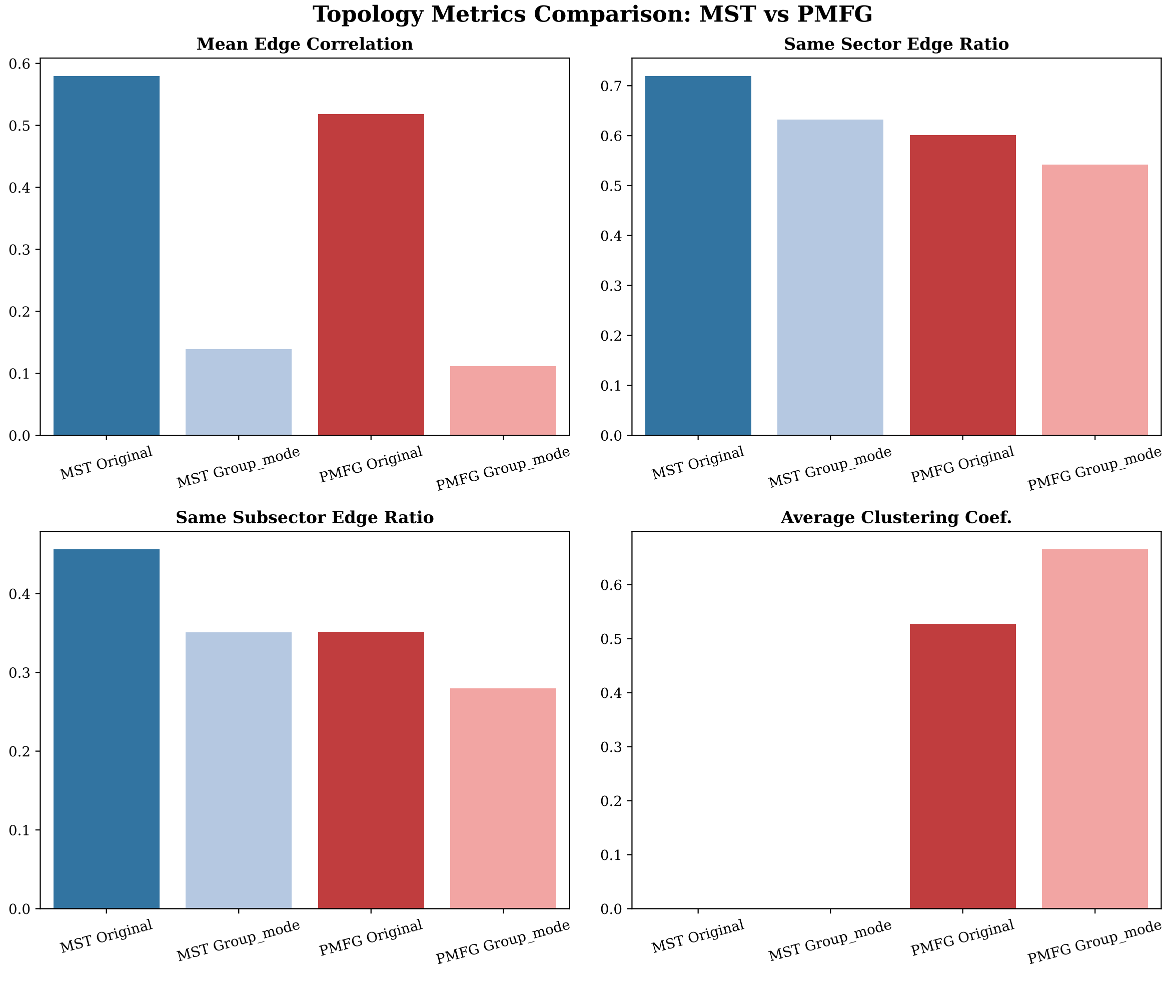


Figure 13: Network topology comparison across original, filtered and group-mode MST/PMFG representations.

8.5. Hub reorganization after RMT filtering

Hub rankings summarize which assets occupy central positions in the constructed networks. Figure 14 compares hub rankings before and after filtering. In the original PMFG, GGBR4 is the top betweenness node and BBDC4 is the top degree node. In the group-mode PMFG, GUAR3 becomes the leading node by both degree and betweenness.

This change in hub ranking is informative for interpretation. Original hubs can reflect market-wide correlation strength and broad exposure to the common component. Group-mode hubs are interpreted as localized bridges in the network obtained after removing the leading common component. The appearance of assets such as GUAR3, BBDC3, VALE3, CCRO3, SAPR4 and CMIG3 in group-mode rankings suggests that centrality should be interpreted conditionally on the matrix used to construct the network. Because hub identities can be sensitive to small changes in edge ranking, these rankings are interpreted as descriptive summaries rather than stable structural labels.

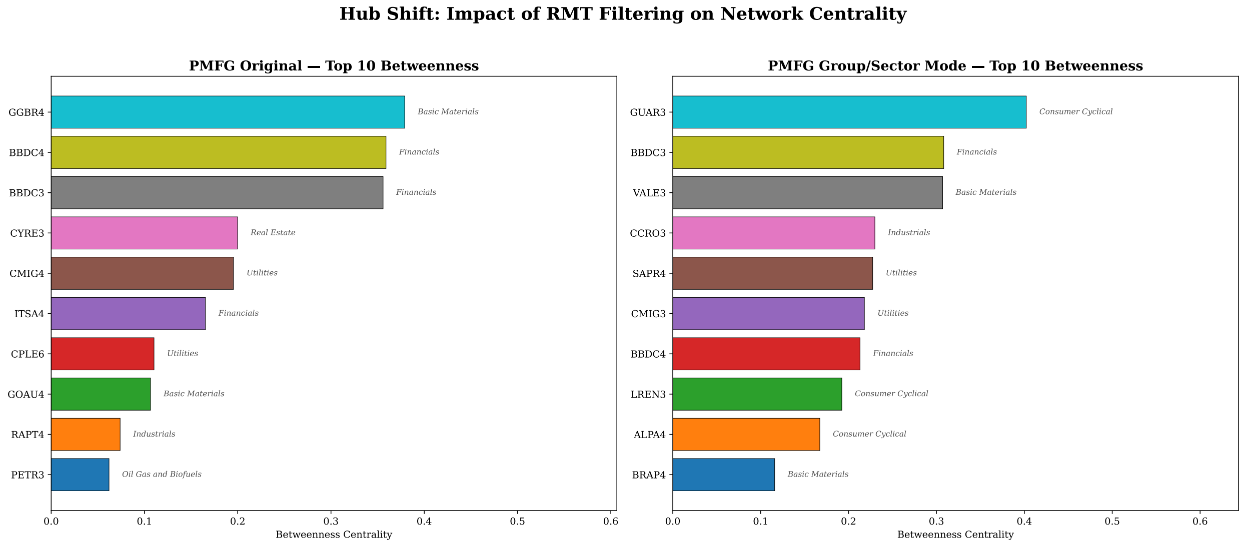


Figure 14: Hub-rank comparison across original and group-mode networks. RMT filtering changes which assets occupy central positions in the dependency topology.

9. Aggregated subsector dependency network

Asset-level networks can be difficult to interpret when many share classes and sectors are present. The aggregated subsector dependency network converts ticker-level dependencies into a mesoscopic map. Nodes represent subsectors and edges summarize average retained dependency strength between subsectors. This is analogous to sector or country aggregation maps used in the financial dependency literature, but adapted to the Brazilian equity universe [33].

Figure 15 presents the resulting network. The network contains 22 subsectors and 66 edges, with density 0.2857. The mean edge dependency is 0.0566 and the median edge dependency is 0.0561. These values are lower than typical raw asset-level correlations, which is expected because the network summarizes filtered and aggregated dependencies rather than direct pairwise asset correlations.

The top-degree subsector is Retail, indicating broad direct connectivity in the aggregated dependency map. Electric Utility has the highest betweenness, suggesting that it occupies a bridge-like position in the mesoscopic topology. Apparel and Footwear has the highest weighted degree, reflecting stronger dependency intensity across its retained connections. These summaries help translate the ticker-level PMFG topology into economic language.

The subsector network also provides a bridge to forecasting. If network topology contains information beyond univariate volatility persistence, then node-level or group-level centrality measures may help explain future volatility regimes. The forecasting section tests this idea in a conservative first experiment.

Table 9: Aggregated subsector network summary.

Subsectors	Edges	Density	Mean edge dep.	Median edge dep.	Top degree	Top betweenness	Top weighted degree
22	66	0.2857	0.0566	0.0561	Retail	Electric Utility	Apparel and Footwear

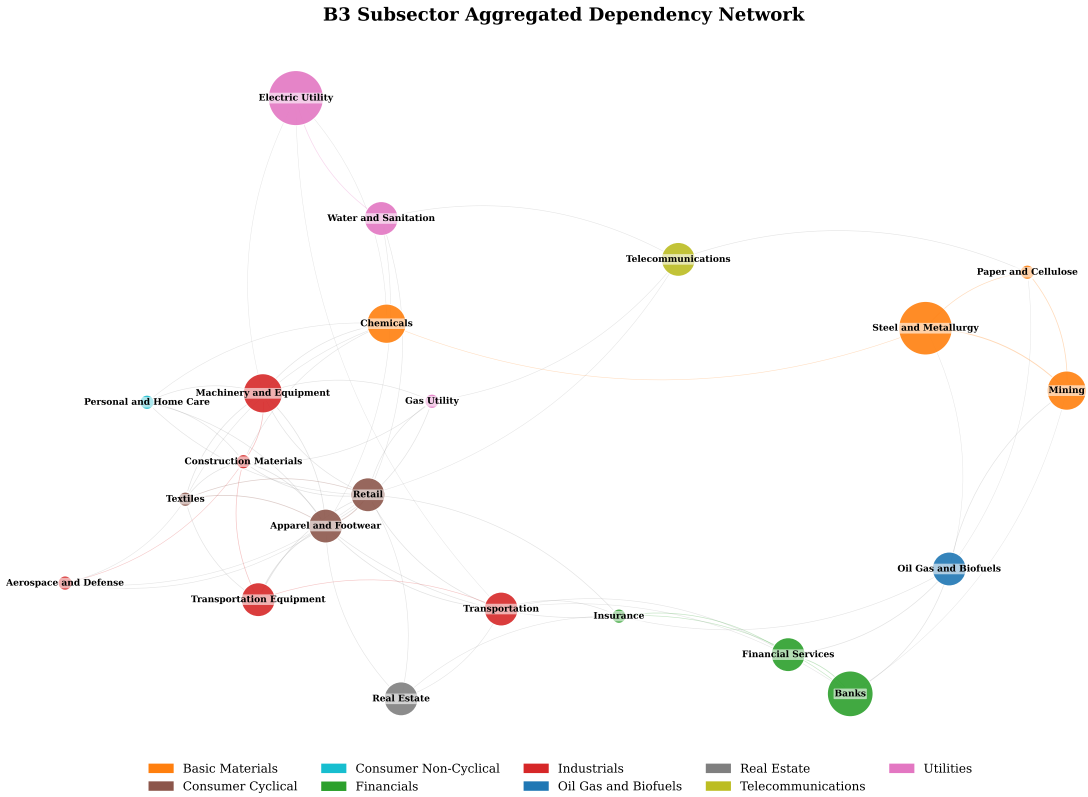


Figure 15: Aggregated subsector dependency network. Asset-level dependencies are summarized into subsector-level connections, providing a mesoscopic view of the Brazilian equity market.

10. Volatility forecasting

10.1. Forecasting dataset and temporal split

The forecasting experiment evaluates whether RMT and network features contain incremental information for realized volatility. The experiment is deliberately conservative. It uses PETR4, VALE3 and BBDC4 as target assets and compares network-augmented machine-learning models with strong econometric baselines. The data are split temporally: training covers 2006–2017, validation covers 2018–2020 and testing covers 2021–2025. This temporal split is designed to reduce look-ahead bias and to evaluate model selection on observations later than those used for training. Because some network and RMT features are based on static or slowly changing structures, the forecasting experiment is interpreted as an incremental-information test rather than as a fully real-time trading design.

10.2. GARCH persistence and econometric benchmarks

The GARCH(1,1) estimates with Student-t innovations indicate high volatility persistence. Table 10 reports ω, α, β and $\alpha+\beta$. The persistence values are 0.9911 for PETR4, 0.9946 for VALE3 and 0.9828 for BBDC4. All three estimates satisfy $\alpha + \beta < 1$, but the values are close to unity, indicating highly persistent conditional variance dynamics. These high values help explain why classical volatility benchmarks are difficult to outperform and are consistent with the slow decay of absolute-return autocorrelations documented in Section 5.

Table 10: GARCH(1,1) persistence estimates for the forecasting assets.

Asset	ω	α	β	$\alpha+\beta$
BBDC4	0.0729	0.0520	0.9308	0.9828
PETR4	0.0780	0.0758	0.9153	0.9911
VALE3	0.0417	0.0512	0.9434	0.9946

10.3. Machine-learning feature sets

The machine-learning models use three nested feature sets in an ablation design. Set A contains classical volatility features. Set B adds market and RMT variables. Set C adds MST and PMFG network variables. This nesting allows the experiment to assess whether network topology adds incremental information beyond classical lagged-volatility persistence and broad market/RMT features.

10.4. Out-of-sample model comparison

Figure 16 summarizes out-of-sample QLIKE loss. Lower QLIKE is better. For machine-learning models, the reported feature set is selected using the validation period and then evaluated on the test period.

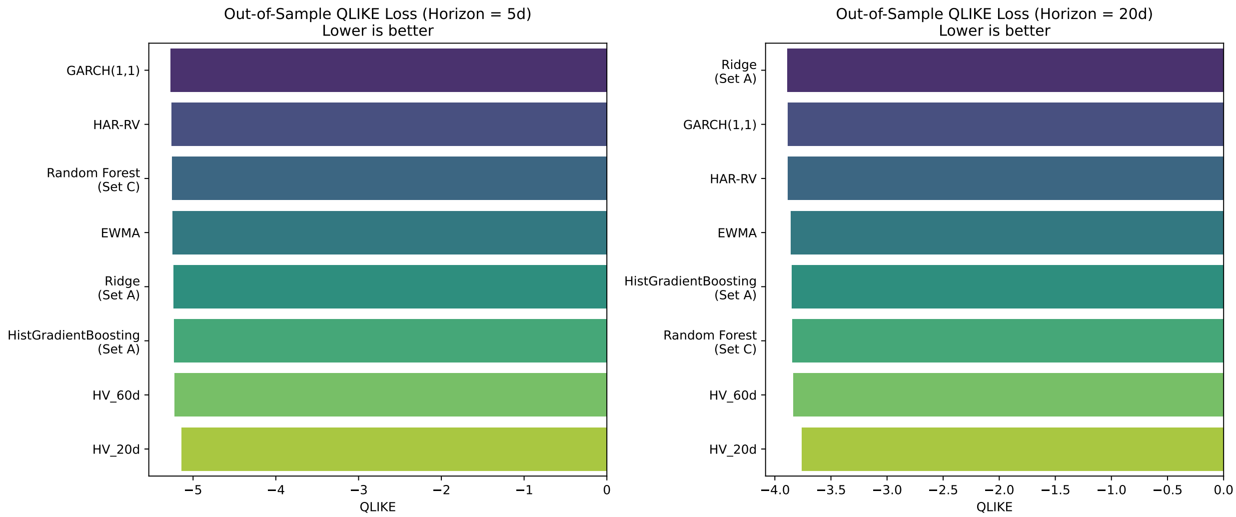


Figure 16: Volatility forecasting comparison for econometric and machine-learning models. Machine-learning labels show the best feature set selected for each model and horizon.

At the 5-day horizon, GARCH(1,1), HAR-RV and Random Forest are highly competitive. GARCH has average test QLIKE close to -5.27, HAR-RV is close to -5.26 and Random Forest with Set C is close to -5.25. The differences are small; therefore, the appropriate interpretation is not that machine learning dominates, but that the network-augmented Random Forest reaches performance comparable to strong classical benchmarks.

At the 20-day horizon, Ridge models with Set A or Set B obtain slightly lower average QLIKE than GARCH in this experiment. Ridge Set A and Set B are close to -3.89, while GARCH is close to -3.88. No formal equal-predictive-ability test is reported, so these differences should be interpreted descriptively. Random Forest Set C remains competitive but does not dominate. This result suggests that longer-horizon volatility may be easier to forecast with stable linear shrinkage features, while nonlinear models require careful regularization and validation.

Table 11: Aggregated test-set forecasting comparison (2021–2025). Values are averaged across PETR4, VALE3 and BBDC4.

Horizon	Model	Feature set	MAE	RMSE	QLIKE	R^2_{oos}
5	GARCH(1,1)	Classical	0.0156	0.0213	−5.2700	0.1600
5	HAR-RV	Classical	0.0148	0.0198	−5.2600	0.2200
5	Random Forest	Set C (+Network)	0.0151	0.0204	−5.2500	0.2000
5	EWMA	Classical	0.0153	0.0207	−5.2500	0.1900
20	Ridge	Set A (Classical)	0.0241	0.0328	−3.8900	0.3900
20	Ridge	Set B (+Market/RMT)	0.0242	0.0329	−3.8900	0.3900
20	GARCH(1,1)	Classical	0.0278	0.0381	−3.8800	0.2300
20	HAR-RV	Classical	0.0260	0.0356	−3.8800	0.2800
20	Random Forest	Set C (+Network)	0.0287	0.0393	−3.8500	0.1700

10.5. Realized vs predicted volatility diagnostics

Aggregate metrics can hide important time-series behaviour. Figure 17 compares realized 20-day volatility with GARCH and Random Forest forecasts for PETR4, VALE3 and BBDC4. This diagnostic is included in the main text because it illustrates how forecasts track changing volatility regimes.

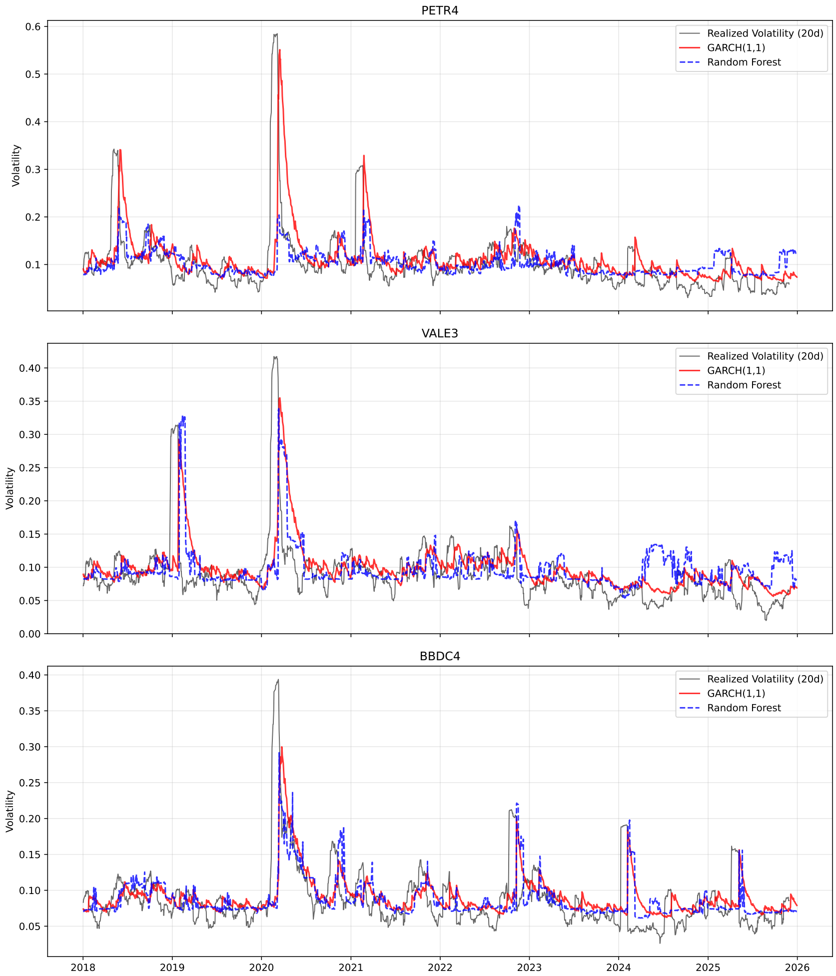


Figure 17: Realized versus predicted 20-day volatility for selected models and assets. The plot illustrates regime tracking and forecast smoothness.

The diagnostic plots suggest that broad volatility regimes are tracked reasonably well, but extreme spikes remain difficult to forecast. GARCH forecasts respond through the persistence of conditional variance, while Random Forest forecasts can appear smoother or more step-like because tree-based predictions are piecewise constant over feature regions. This behaviour is consistent with the aggregate results: nonlinear models can be competitive, but lagged volatility and volatility persistence remain the main sources of predictive information.

10.6. Feature importance and network signal

Random Forest feature importance provides a model-specific diagnostic of which predictors contribute most to the fitted model. Figure 18 shows that classical volatility features dominate the model. Rolling absolute returns, lagged realized volatility, rolling volatility and EWMA variables carry most of the importance.

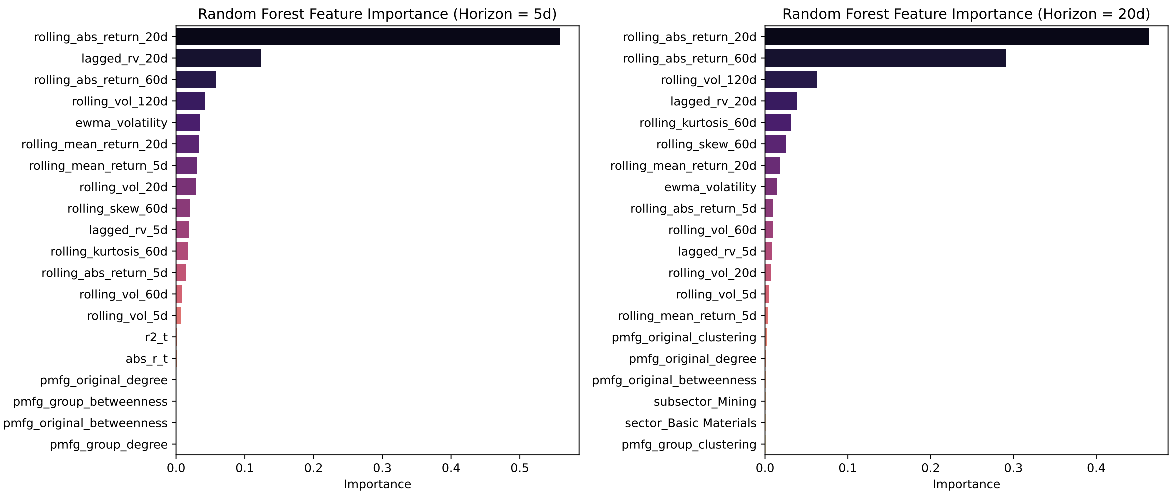


Figure 18: Feature importance for Random Forest volatility forecasting models. Classical volatility predictors dominate the selected rankings, while selected PMFG-based and sector/subsector variables appear with smaller but nonzero importance in the network-augmented feature set.

The strongest features are rolling absolute return over 20 days, lagged realized volatility and longer rolling volatility measures. Network features have smaller importance values than classical volatility predictors, but selected PMFG variables enter the top-ranked predictors, especially at the 20-day horizon. Because Random Forest importance measures can be affected by correlation among predictors, these rankings should be interpreted as model diagnostics rather than causal measures of variable relevance. The appropriate conclusion is therefore incremental: topology does not replace classical volatility features, but it may add structural information when included in a broader feature set.

11. Discussion

Taken together, the empirical results support a layered interpretation of Brazilian equity dependencies, consistent with the multi-scale perspective advocated by Raddant and Di Matteo [33]. The first layer is a market-mode component. It is reflected in the moderate positive average correlation ($\bar{\rho} \approx 0.24$), the increase in rolling average correlation around stress periods and the very large leading eigenvalue ($\lambda_1 = 21.65$, approximately 37.3% of the trace of the correlation matrix). This component is economically interpretable as broad market synchronization. In the Brazilian context, such synchronization may reflect common exposure to domestic monetary conditions, exchange-rate dynamics, country risk and global commodity cycles, although these channels are not separately identified in the present analysis. The component is also methodologically important because it can dominate raw networks and obscure weaker local structures.

The second layer consists of candidate group and sector structure. Within-sector correlations are higher than between-sector correlations on average, and the RMT decomposition identifies four additional eigenvalues above the random upper edge. Eigenvector loadings and group-mode matrices suggest that commodity, financial, telecommunications and utility-related structures contribute to the non-random part of the spectrum. This interpretation is descriptive rather than causal: the analysis identifies structure in the dependency matrix, not the economic mechanisms that generate each edge.

The third layer is noise-compatible residual dependence. Not every correlation is economically meaningful, and not every edge retained by a graph should be interpreted as a stable economic relationship. This is why the paper combines RMT, heatmaps, dendrograms and network metrics. A result is more credible when it appears consistently across several representations rather than only in one visualization.

The network results demonstrate that market-mode removal changes topology in a structured way. Original networks have stronger average edge correlations, while group-mode networks have much lower average edge correlations. However, the group-mode PMFG has higher clustering (0.6653 versus 0.5273). This combination is a central finding: lower average edge strength does not mean absence of structure. It means that the dominant common component has been removed and localized mesoscopic organization becomes easier to observe [40, 41]. The PMFG is particularly suited for this analysis because it preserves richer topology than the MST, retaining 168 edges with 166 triangles and 55 4-cliques that allow hub-rank shifts, clustering changes and subsector dependency maps to be studied in detail.

Clustering summarizes hierarchical organization, but it does not identify graph-theoretic hubs, bridges or local cycles. The transition from dendrograms to filtered financial networks provides complementary information. The hub reorganization after RMT filtering is particularly informative: original hubs such as GGBR4 and BBDC4 are central in networks that include the common component, while group-mode hubs such as GUAR3 occupy central positions after the leading component is removed. This suggests that centrality in financial networks should be interpreted conditionally on the matrix used to construct the network.

The forecasting results should be interpreted with appropriate caution. They do not show that machine learning uniformly dominates classical volatility models. Instead, they show that network-augmented models can approach strong baselines and that selected PMFG variables have nonzero importance in Random Forest models. The persistence of GARCH(1,1), with $\alpha + \beta$ consistently above 0.98, indicates that autoregressive volatility dynamics remain the dominant forecasting signal for Brazilian equities. Network features complement rather than replace this signal. The appropriate interpretation is incremental: topology adds structural information from the cross-sectional geometry of the market to the temporal dynamics of individual asset variance.

The broader contribution is methodological. The paper connects econophysics filtering, financial networks and volatility forecasting in a single reproducible pipeline for B3 equities. By documenting each step transparently and evaluating network features against demanding econometric benchmarks, the analysis provides a template that can be extended to other emerging markets, different time scales and more sophisticated forecasting architectures.

12. Limitations

The first limitation concerns time variation. The RMT and network features used in the initial forecasting exercise are based on full-sample or slowly changing structures. They are useful for understanding the market, but they should not be interpreted as fully real-time predictors unless reconstructed recursively using only information available at each forecast origin. A stricter forecasting design should use rolling RMT and rolling network features.

The second limitation concerns the target set. The forecasting experiment uses PETR4, VALE3 and BBDC4. These assets are liquid and economically important, but they do not cover the full B3 cross-section. Future work should extend the forecasting exercise to a larger set of assets and evaluate whether network features help more for some sectors than others.

The third limitation concerns data quality. The core historical universe is sufficiently clean for the current analysis, but broader or more liquid modern universes may contain extreme adjusted-return events that require additional validation. These events may reflect corporate actions, ticker transitions, units or adjustment-factor issues. They should be flagged, inspected and either corrected or handled through robustness checks before broader universes are used for final network claims.

The fourth limitation concerns sector classification. The macro-sector and subsector map is an initial classification. It is adequate for testing whether within-sector correlations exceed between-sector correlations, but it should be reviewed ticker by ticker before submission. This is especially important for holding companies, units, firms with multiple share classes and companies with mixed exposure across sectors.

Finally, the machine-learning analysis is limited to three target assets and static network features. Feature importance in tree models is informative but does not establish causal mechanisms. The nonzero importance of PMFG features suggests incremental signal, but the magnitude is small compared with classical volatility predictors. Future work should include rolling network features, DCC-GARCH comparisons, a broader target asset set, temporal neural architectures and graph neural networks.

13. Conclusion

This paper develops an econophysics pipeline for analysing the dependency structure of Brazilian equities traded on B3 from 1998 to 2025. The main findings can be summarized along four dimensions.

First, the stylized-fact and correlation analysis provides descriptive evidence that Brazilian equities display empirical signatures commonly associated with complex financial systems: heavy-tailed return behaviour, volatility clustering, persistent absolute-return dependence and a moderate but positive average cross-correlation ($\bar{\rho} \approx 0.24$). Within-sector correlations are larger than between-sector correlations on average, supporting the interpretation that sectoral organization is reflected in the empirical dependency matrix.

Second, Random Matrix Theory provides a spectral decomposition of the correlation matrix. Five eigenvalues exceed the Marčenko–Pastur upper bound, with the largest eigenvalue accounting for approximately 37.3% of the trace of the correlation matrix. The leading component is interpreted as a market-mode component, while the remaining eigenvectors outside the random band are consistent with commodity, financial and utility-related group structures. The reconstruction diagnostics indicate numerical accuracy within floating-point precision, and the group-mode matrix retains localized dependency patterns after the dominant market component is removed.

Third, financial network analysis through MST and PMFG representations shows that RMT filtering changes the topology of asset dependencies. The group-mode PMFG has lower mean edge correlation but higher clustering than the original PMFG, indicating stronger local clustering after market-mode removal. Hub rankings shift from networks dominated by broad market co-movement to networks where different assets occupy central positions after excluding the leading component. The PMFG retains 168 edges with triangles and 4-cliques, providing local topological structures that are absent from the MST and useful for mesoscopic analysis.

Fourth, the volatility forecasting experiment provides preliminary evidence that network topology can complement classical volatility predictors. GARCH(1,1) and HAR-RV remain strong benchmarks, especially at short horizons, consistent with the high persistence ($\alpha + \beta > 0.98$) observed in all three target assets. Machine-learning models with PMFG-derived features approach these baselines, and at the 20-day horizon Ridge Regression obtains slightly lower average QLIKE than GARCH in this experiment. Since no formal equal-predictive-ability test is reported, this forecasting result should be interpreted descriptively. Feature importance analysis shows that classical autoregressive volatility features dominate, while selected PMFG centralities contribute smaller but nonzero predictive information.

The appropriate conclusion is incremental: network features do not replace classical volatility dynamics, but they add structural information from the cross-sectional geometry of the market. Future work should extend this framework by constructing rolling RMT and network features for real-time forecasting, expanding the target asset set, comparing against multivariate volatility models such as DCC-GARCH and evaluating temporal neural network and graph neural network architectures.

A. Supplementary figures

Additional figures and tables generated by the computational pipeline are retained as supplementary material. These include asset-universe tables, return summaries, correlation matrices, RMT eigenvector loadings, network edge lists, clique lists, centrality rankings and detailed forecasting results by asset, horizon, model and feature set.

B. Supplementary tables

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The empirical pipeline uses locally processed B3/BovDBv2 daily adjusted price records. The public BovDBv2 dataset is available through Harvard Dataverse under DOI `10.7910/DVN/TMB4IG`. Derived scripts and reproducibility materials will be made available in a public repository upon publication, subject to redistribution constraints for raw market data.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT for language refinement, structural organization and code-review support. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Andrey V. S. Souza: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. **Bruno L. Dalmazo:** Methodology, Supervision, Writing - review and editing. **Giancarlo Lucca:** Methodology, Validation, Writing - review and editing. **Eduardo N. Borges:** Supervision, Methodology, Writing - review and editing. **Richard F. Pinto:** Software, Data curation, Validation. **Fabian C. Cardoso:** Data curation, Resources, Writing - review and editing. **Viviane L. D. de Mattos:** Formal analysis, Supervision, Writing - review and editing. **Rafael A. Berri:** Supervision, Project administration, Writing - review and editing.

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